

Customary Tenure and Agricultural Investment in Uganda

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Abstract

Customary tenure in Uganda often features land rights being held by different individuals in a community, creating complex incentives for long-term agricultural investment. I introduce a theoretical model of these incentives, exploring farmers' strategic responses to changing land pressures, which may change the social determinants of tenure insecurity. Empirically, I find results consistent with the model in Uganda. As land pressures rise, long-term input use responds more on freehold than on customary parcels. This can be explained by elite incentives for expropriation, which intensify with external demand for land, creating insecurity on customary parcels. This study offers a nuanced conceptualization of the relationship between tenure systems and agricultural investment, incorporating social dynamics (such as elite expropriation) in both theory and empirics.

1 Introduction

Secure property rights are fundamental to investment and prosperity, and land tenure in particular is critical for poor farmers to make long-term investments and improve agricultural productivity (Place, 2009; Aberra and Chemin, 2021). In much of Sub-Saharan Africa, customary tenure, a socially-created and enforced, typically undocumented traditional form of land rights, predominates (Alden Wily, 2011). Economists have traditionally focused on the undocumented and therefore theoretically less secure nature of customary tenure when compared with freehold land, where the owner enjoys all rights in perpetuity (Fenske, 2011). Large-scale

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formalization policies across much of the continent have aimed to address this insecurity to allow farmers to make transformative investments in their own farms (Ali et al., 2016; Lawry et al., 2023). However, empirical work has been more mixed than theory; customary land is not always less secure in farmers’ own estimation (nor under-invested relative to freehold land), and formalization has not always unlocked investment (Bambio and Bouayad Agha, 2018; Lawry et al., 2017; Fenske, 2011; Higgins et al., 2018; Boxho et al., 2022; Bellemare, 2013). By contrast, qualitative work has emphasized the socially-constructed nature of customary tenure, where multiple individuals may hold overlapping rights over a single piece of land, and in particular, local elites such as chiefs or lineage heads hold transfer rights (Cotula et al., 2007; Ensminger, 1997; Alden Wily, 2011; Van Leeuwen, 2014; Barry and Danso, 2014).¹ The distribution of rights can therefore influence the (perceived) security of tenure (Goldstein and Udry, 2008), as other rightsholders may claim the land, although the structure of rights are distinct from the security of those rights (Le Rossignol et al., 2024). Often framed around issues of elite capture, these patterns have been difficult to document in quantitative survey data, although their importance is highlighted in qualitative work on land disputes and dispossession (Kandel, 2022).²

I incorporate the social dynamics of land rights, where multiple rightsholders interact strategically, into a theoretical model of agricultural investment and empirical tests using data from Uganda. My model’s primary contribution lies in capturing not just different investment incentives under different tenure regimes, but how these incentives change in response to external land pressures. As land pressures increase, the relative output price of (input-intensive) cash crops should increase, increasing investment, but local elites may face more incentive to expropriate customary land for sale to outsiders (Cotula et al., 2007; Ubink, 2008; Berry, 2018; Bambio and Bouayad Agha, 2018).³ If farmers begin to worry about elite expropriation,⁴ this should impact their investment decisions. In particular, I find that farmers under-invest in long-term

¹A hallmark of the customary is its very localized, negotiated nature; this enables its adaptability to changing circumstances but can also marginalize those in weaker bargaining positions (Cotula et al., 2007; Lawry et al., 2023; Chimhowu, 2019). Ensminger (1997) argues that “a common characteristic in almost all African customary systems is for use rights to be assigned at the household level, whereas transfer rights are assigned at a higher level such as the lineage, clan, or chiefdom.” This means that although individuals may be quite secure in their ability to use a particular plot of land in their lifetime, adaptive reallocation within the community could occur in response to changing circumstances. Customary leaders also are able to allocate (often unused) land to migrants into an area as well as adjudicate any conflicts that may occur between members of the community. Although the particular ways these rights are divided may vary considerably across the region (Hasanbasri et al., 2022), this ‘dis-bundling’ of property rights is common and adds a strategic dimension to any decisions about land. At the same time, however, the dis-bundling does not closely mirror a sharecropping contract, as the elite holding transfer rights does not hold residual profits and there is no formal contract between multiple rightsholders; a classic principal-agent model with potential holdup problems is therefore of limited relevance (Aghion and Tirole, 1997; Goetz et al., 2019).

²This more localized dispossession is distinct from the large scale land acquisitions that have received international attention, although the phrase ‘land grabs’ is used as a catch-all term (Van Leeuwen et al., 2023).

³Kandel (2022) documents that those with off-farm salaried income are “increasingly acquiring rural land in order to invest in small-scale commercial agriculture or as speculative investments,” a phenomenon termed ‘emergent farmers’ by Sitko and Jayne (2014). Indeed, Lawry et al. (2023) describes greater demand for land in response to population growth as one of the primary threats to customary tenure systems. This ability for local land elites to take advantage of new economic situations and their better connections to outsiders has been documented in other contexts, including Pakistan (Beg, 2021).

⁴Traveling around Uganda, it is common to see signs on properties especially agricultural land stating “this land is not for sale”: an indication that farmers are indeed worried about elite expropriation for sale to outsiders.

agricultural inputs on customary parcels (when compared with freehold) as land pressures rise.

I begin with an endogenous model of customary tenure and agricultural investment where farmers can make socially-recognized investments in their land to enhance their tenure security, with inputs applied to both staple and cash crops. Long-term investments are weakly lower on (relatively insecure) customary land than on freehold land, although the customary system may provide sufficient investment incentives (Brasselle et al., 2002; Lawry et al., 2023). This connects with empirical work linking customary tenure arrangements to a (lack of) adoption of cash crops, including Deininger et al. (2021)⁵ and Goldstein et al. (2018)⁶. New pressures on land values, which often emerge as a result of increased access to urban centers, also raise the relative price of input-intensive cash crops. This implies that optimal investment is an (increasing) function of land pressures on freehold land. However, I explicitly consider a case common in customary tenure systems: one household or individual may hold primary use rights, while another, such as a lineage elder, may hold rights of transfer (Ensminger, 1997; Cotula et al., 2007).⁷ This partition of rights means that outsiders may struggle to identify land ‘owners’, finding local tenure systems illegible (difficult to parse without deep knowledge of the area and people nor the socially-recognized investments that demonstrate and secure tenure locally). Elites are then able to sell land used by other members of the community for their own private benefit (Ubink, 2008; Sitko et al., 2014). Their incentives to do so are greater in contexts with rapidly increasing land values (Barry and Danso, 2014; Cotula et al., 2007). This means that rising land values, driven by sales options to outsiders, may lead local elites to assert their historic right to sell land with little or no advance notice, creating insecurity for smallholders (Perego, 2019). This expropriation is a real threat for Ugandans: the Prindex survey of Uganda finds that 22.4% of respondents report personally experiencing any involuntary land loss (Prindex, 2020), and qualitative work finds that dispossession is more likely for the poor and socially vulnerable (Kandel, 2022).⁸ The model predicts that the farmer, anticipating elite expropriation, may actually feel more tenure insecure and therefore make fewer long-term investments on customary land as land values rise, in contrast to the freehold case.

In order to empirically test the implications of this model, I use data from a survey of 2,189 farmer households in Uganda. My primary empirical specification looks at heterogeneous responses to rising land pressures between customary and freehold land for a variety of investments of long and short duration.

⁵Who find that bequest and sales rights, particularly those held by men, are strongly related to cash crop adoption; these long-term rights are more important than short-term decisionmaking rights.

⁶Who find that households who received demarcation of their customary parcels in Benin were 23-48 percent more likely to grow perennial cash crops.

⁷I focus on these two rights given their first-order importance for investment incentives, despite the plethora of other secondary rights that may be held by various individuals.

⁸The public Prindex data does not allow us to empirically examine where these expropriations happen, however, nor the kind of land which was expropriated. Nevertheless, it is a valuable resource for understanding the prevalence of tenure insecurity and land disputes in Uganda: 16.6% of respondents report having experienced a dispute over land in the past, and 26% of the adult population report feeling insecure in their land rights.

In order to capture an exogenous component of land pressures, I use two spatial measures derived from Muller-Crepon (2021). The first is the inverse travel time to Kampala, the national capital and source of much outsider demand for land, which incentivizes elite expropriation but also increases the relative price of input-intensive cash crops. The second is the historical decrease in travel time to Kampala due to new or improved roads, relative to the 2005 travel time. This isolates new incentives to expropriate customary land, which play a role in uncertainty and tenure insecurity. These measures should capture the degree to which outsiders are interested in purchasing local land, and therefore the drivers of elite expropriation, but abstract away from other pathways by which land values impact agricultural investment. I also use estimated per-acre rental prices for parcels; with this measure, I control for farmer group fixed-effects to account for unobserved farmer quality characteristics.⁹

Empirically, I find that long-term input use responds to rising land values more strongly on freehold land than on customary land. Elite incentives to expropriate rise along with land pressures, causing tenure insecurity on customary land (seen empirically), and this attenuates the impacts of a shift to input-intensive cash crops. This divergence in input use between freehold and customary parcels is smaller for short-term inputs, perhaps due to additional demands for cash crops or frictions in land and labor markets (Adamopoulos et al., 2022). Nevertheless, the difference in responses is stronger for long-term inputs, which can be attributed to the tenure-security effect. Additionally, the divergence in investment is more responsive to rising land pressures closer to Kampala.

By incorporating the social nature of customary land rights in a model of agricultural investment, this paper avoids the common conflation of incompleteness of rights with insecurity (Doss and Meinzen-Dick, 2020; Hasanbasri et al., 2022; Place, 2009; Le Rossignol et al., 2024).¹⁰ Instead, it considers how multiple rights-holders interact in different land value environments to make more nuanced predictions about investment on customary land, reconciling the mixed evidence in the literature (Fenske, 2011). In addition to bringing qualitative insights to an economic framework, I use economic modeling to explore how farmers would strategically respond to changing incentives. This model captures how the potential for elite capture affects individual farmers, similarly to Aberra and Chemin (2021). It thereby returns the focus to the welfare of

⁹These farmer groups were part of the sampling strategy in the data collection process. They extend the intuition of household fixed-effects to control for unobserved heterogeneity as farmer groups are self-selected, but allow for sufficient identifying variation. A household fixed effects specification looking at differences across tenure types (particularly in West Africa) may identify from households with the greatest differences in tenure security (and thus households most likely to invest differently across plots), as only those households who feel insecure about land allocated through customary mechanisms will purchase other plots (Fenske, 2011).

¹⁰‘Security’ could be composed of multiple elements of duration of tenure, assurance, and completeness of rights. Despite the theoretic clarity of tenure security, “there is no agreed upon way to measure tenure security and results may be related to choice of proxy” (Place, 2009). For example, Bellemare (2013) finds in Madagascar that one measure of tenure security, the presence of title, seems unimportant for investment, while beliefs about rights do matter, while Deininger et al. (2021) in matrilineal areas of Malawi that rights to bequeath and sell land are more important than short-term management rights. Gumucio et al. (2025) differentiates robustness, completeness, and durability of rights; in this paper, I consider how changing external circumstances such as land pressures can shape the robustness of customary rights, and thereby reduce their durability.

smallholders, and by examining a broader population allows me to quantitatively document the effects of (potential) elite capture (which historically has been difficult due to the rarity of observed cases). Unlike many previous papers which have conceptually modeled the impacts of tenure *security* but then empirically used tenure *type* as a (poor) proxy, I consider the particular institutional arrangements of tenure type and the incentives it creates throughout my model, therefore linking more closely to my empirical tests (Fenske, 2011; Abdulai et al., 2011; Linkow, 2019). Uganda’s tenure context, where customary and freehold parcels exist in the same local areas due to exogenous historical factors (Deininger and Ali, 2008), also allows me to isolate the impact of land pressures on tenure insecurity from structural issues such as land and labor market frictions leading to misallocation.¹¹ These potentially missing markets are common across parcels in a given location, and therefore are unlikely to drive divergence between parcels of different tenure types in an area.

The remainder of this paper consists of a description of the context of customary tenure and agricultural investment in Sub-Saharan Africa, with a particular focus on Uganda, followed by a model which captures many of the insights explored. I then discuss the data used, as well as an empirical strategy with several hypotheses laid out. I present results for a variety of short- and long-term inputs, and conclude by discussing their implications for the model’s relevance to the Ugandan context.

2 Context

Importantly, the dispersion of rights in many customary tenure systems across the continent is not in itself a cause of instability or under-investment: “sufficient investment incentives tend to be provided by basic rights of use that, under normal circumstances, are guaranteed to many villagers (including migrants) by the local informal order” (Brasselle et al., 2002). Within a local community, farmers are able to demonstrate their stewardship of the land via socially-recognized investments, such as planting trees, that signal socially that the land is being used (Awanyo, 2009). However, the continual adaptation of customary institutions means that such systems are dynamic and see considerable changes over time. My model explores how outside, market-based pressures on land shape customary tenure security and investment incentives.

In particular, I focus on the role of elites in customary tenure. A common feature of customary systems across Uganda and the continent is that elites, representing the family or lineage writ large, serve as trustees or administrators of land owned by the family group (Ensminger, 1997). These elites may be lineage heads,

¹¹The restrictions on land transferability, as in Adamopoulos et al. (2022) and Adamopoulos and Restuccia (2019), or the sectoral misallocation of labor, as in Adamopoulos et al. (2024), do lead to underinvestment in the land and lower agricultural productivity, and could explain some of the gap I observe for all investments between customary and freehold parcels. However, these market frictions do not explain a differential gap between short and long-term inputs, nor that differential gap widening with increasing land pressures, which I document and which is predicted by my model as stemming from insecurity.

(land) chiefs, elders, or others in positions of social power. Often, they hold transfer rights in order to (re)allocate land within the community in response to shocks (such as the death of a large landholder without heirs or an influx of new migrants to an area), as well as to prevent the alienation of a communal resource without the consent of the community.

However, beginning with colonial administrators, outsiders have attempted to fit notions of ownership based on freehold tenure onto customary tenure systems, and this institutional mismatch has created ambiguity about who the ‘owners’ of land actually are. This ambiguity can be exploited by local elites: “There is a fine line between chiefs as (often self-declared) owners of all land in customary laws, and chiefs as trustee administrators” (Alden Wily, 2011). Elites are able to respond as individuals rather than as guardians of the corporate group, and can thus assert their rights to outsiders unfamiliar with the complexities of local tenure arrangements.

The qualitative literature on customary tenure has documented the tensions and conflicts that emerge as customary systems adapt to external pressures such as rising land values. Most cases have considered land values rising due to *non-agricultural* uses, such as urban and peri-urban expansion, or natural resource extraction; this also includes the ‘emergent farmers’ documented by Sitko and Jayne (2014) who are more commercially-oriented but no more productive than smallholders. Some large scale land acquisitions have been for agricultural purposes, although generally requiring scale or technologies such that local smallholders are unable to participate. Therefore, this should be thought of as an increase in the *marketable* value of land rather than its value to smallholder agriculture. “As land values rise, farmers may be forced or tempted to sell their land. Where land is still under customary chiefs, these may be tempted to sell off lands ... regardless of the views of those actually farming this land” (Cotula et al., 2007). In Ghana, “many of these multiple sales are by different people in a family lineage, each contending that they have the status to sell under the customary system” (Barry and Danso, 2014); this reinterpretation of customary law is also documented in Niger by Turner and Moumouni (2019). Despite abundant stories of how “local elites have been able to use their position and the ambiguities of customary law to appropriate land to further their own economic and political interests” (Ubink, 2008), especially in peri-urban areas but also in many rural ones (Ubink and Quan, 2008), this particular facet of how customary tenure adapts to external pressures has been little studied by economists, with the notable exception of Berry (2018)’s work on land governance in these areas.

2.1 Land Rights in Uganda

Given the enormous diversity of customary tenure arrangements throughout Sub-Saharan Africa, it is worth delving into the particular case of Uganda. Between 12-20% of land is subject to formal title, so customary

tenure remains significant throughout the country (Alden Wily, 2011; Goldstein et al., 2016), and customary law is recognized under the Constitution (Foote, 2025).¹² Customary authorities still play a major role in Uganda: Gochberg (2021) argues individuals may choose not to title their land given the social costs of leaving the customary system (echoing Honig (2022)’s work in other countries on the continent).¹³ This customary land is subject to the kind of elite expropriation I model: Adoko and Akin (2011) frames the likelihood of elite land grabbing as a function of physical or ‘social strength’ in the community and family, and Gochberg (2021)’s qualitative work relates elites can “go so far as to survey another individual’s land at night” (p. 10).

Even within Uganda, there is some variation in the relative strength and allocation of bundles of rights. In the west, where population pressures on the land are higher, there is more individualized customary tenure, while the relatively land-abundant north retains stronger rights for customary authorities (Van Leeuwen, 2014), in part due to historical agropastoral systems and collective rights (although post-war resettlement and return complicates customary systems in the north (Foote, 2025; Sjögren, 2014) and southwest (Mathenge, 2013)). In the historic Kingdom of Buganda, in the center of the country, the *mailo* system is key, a remnant of British indirect colonial rule.¹⁴ *Mailo* land only exists in the historic kingdom of Buganda, however, in central Uganda, and even there, customary land is still substantial.¹⁵ A given smallholder’s likelihood of having their land expropriated also varies, particularly with their social status within the community (connections to elites, for example, gender, or generation) (Goldstein and Udry, 2008; Cherchi et al., 2019; Poulin and Pierotti, 2024; Linkow, 2019).

Qualitative work has documented throughout the country that rights to the land in perpetuity are differentiated from responsibility for managing the land (Adoko and Akin, 2011). Specific land rights regimes can be incredibly precise: Howard and Nabanoga (2007) document that rights are determined for individual people over individual plants on different types of land under different circumstances.¹⁶ I leave these particu-

¹²This is slightly higher than for the continent as a whole, in part due to colonial administration which focused on documenting land allocation in Uganda.

¹³In lab-in-the-field experiments, references to traditional authorities such as the *kabaka* (the king of Buganda, one of the major regions in Uganda) induce higher contributions to public goods games (Goist and Kern, 2018).

¹⁴Much of the academic literature on land rights in Uganda has looked at the *mailo* system. In 1900, the British signed a treaty with the Kingdom of Buganda which allocated square-mile tracts in the center of the country to Buganda elites for their ‘ownership’ as absentee landlords. These tracts were then sub-leased to the actual inhabitants and land users (Deininger and Castagnini, 2004). Ever since, there has been a significant tension over government policy towards *mailo* owners and tenants and how to balance the two parties’ competing interests (Coldham, 2000). The 1998 Land Act attempted to balance these by giving *mailo* owners the powers of a freehold owner, while still recognizing the rights of ‘lawful occupants’ of the land who had used it for more than 12 years (Coldham, 2000). This guaranteed occupants’ tenure security (including inheritance rights and the ability to sub-let with consent of the owner), while requiring continuous possession of the land and a nominal rent payment. However, *mailo* tenants still face substantial tenure insecurity, particularly regarding expropriation by owners.

¹⁵My survey includes one district with a substantial share of *mailo* land; 20% of parcels there are reported as *mailo*, with less than 1% reported similarly in any other surveyed district. However, a nearly identical share of parcels in that district are reported as under customary tenure. For this analysis, I categorized *mailo* parcels (reported by tenants) as customary land, although results are robust to their exclusion from the analysis entirely (results available upon request).

¹⁶For instance, that a woman who is pregnant or sick is allowed to pick certain medicinal plants from someone else’s river plot, but not from a home garden.

larities to the side, focusing on the first-order rights of land use/management and transfer rights, which have the most direct impacts on agricultural investment. The precise institutional details shape the identities of the different rightsholders in a region (whether Basoga chiefs in the Busoga region in the south-east of the country or Rwots in Acholiland in the north, or other traditional leaders), how elites are selected (hereditary or elected), and in what other domains they have power (whether primarily on land or also other areas such as maintaining social order). Nevertheless, the fundamental distinction I model is common across the country: on customary land, although the farmer has use rights, a customary elite will hold transfer rights.

Deininger and Castagnini (2004)’s examination of land-related disputes in Uganda can illustrate how customary tenure functions and the margins at which it breaks down. They estimate that up to 5% of the rural population is involved in a pending land-related conflict at any given time, mostly between neighbors because boundaries have been exceeded. More recent estimates from Prindex find that 16.6% of the population report ever having experienced a dispute, with 22.4% reporting having personally experienced the involuntary loss of land (Prindex, 2020). The prevalence of boundary disputes has led to many farmers planting trees to demarcate their land; “boundary trees are useful evidence when the land can be visited during a case by customary authorities, but they are less useful when the case is heard in a court of law far away from the land” (Adoko and Akin, 2011). There has been a push to incorporate customary leaders into statutory court systems, with mixed success (Van Leeuwen, 2014).

Historically, proclamations of state ownership of all land under the dictatorship of Idi Amin in 1975 were largely unenforced and ignored by the population (Hunt, 2004). The Land Act of 1998, by contrast, regarded land as the property of the citizens of Uganda and recognized four tenure types: freehold, leasehold, customary, and *mailo* (Joireman, 2007). The law provides for an extensive and decentralized land administration (Tripp, 2004) that could issue titles as well as ‘certificates of customary ownership.’ These certificates registered rights held by multiple people and could, in time, be converted into freehold titles (Coldham, 2000; Perego, 2019). Unlike compulsory registration in other countries in the region, registering rights was voluntary. This leads Coldham (2000) to observe that “where the grant of certificates of title is based on individual applications, there is always a risk of land-grabbing, that is, that an applicant may lay claim to a larger area of land than that to which he is customarily entitled.” Furthermore, the widely-lauded land reform has been largely unfunded and thus little-implemented (Deininger and Castagnini, 2004; Joireman, 2007). Individuals do seem receptive to titling: Cherchi et al. (2019) found in Uganda that 91% of households chose to title their land when offered a fully-subsidized freehold title, and Perego (2019) finds that farmers choose to title their land in response to agricultural price shocks, particularly in areas with greater market access and tenure insecurity.¹⁷

¹⁷Perego (2019) argues that this demand for titling is driven by improved liquidity to pay titling fees from higher crop revenues

Two previous economic analyses have looked at the impacts of tenure on agricultural investment in Uganda. Deininger and Castagnini (2004) focus on land conflicts, but argue that the intensive multi-cropping system prevalent in much of Uganda implies individual crop production functions are inappropriate for productivity measures or investment. Deininger and Ali (2008) use household fixed effects to control for unobservables such as farming skill and find that households invest in both long-term and short-term inputs significantly more on plots they own outright rather than plots they have only customary rights over. Furthermore, they find that registering customary rights (by acquiring a certificate of customary ownership) has little impact on investment, but legal efforts to strengthen occupancy rights do increase investment.¹⁸

3 A model of tenure and investment

Broadly, land tenure has been thought to encourage investment through three possible mechanisms, going back as far as the discussion in Feder (1988, cited in Place (2009)). These have been termed the assurance, collateralizability, and realizability mechanisms: stronger land rights should provide assurance that the farmer will be able to reap gains from investment, increase access to capital by leveraging the land as collateral, and allow the farmer to sell the land and realize gains sooner (as well as transferring land to those most able or likely to invest) (Besley, 1995, cited in Fenske (2011)).

Each of these mechanisms is the subject of an extensive theoretical and empirical literature, and their relevance to customary tenure in Africa has been long debated. Credit markets have been shown to be thin, farmers may be credit rationed for other reasons, and land markets are often missing or face other restrictions (Tabetando et al., 2023; Higgins et al., 2018).¹⁹ Lawry et al. (2017), reviewing a number of studies on property rights interventions, find no consistent evidence on the credit channel. My discussion will focus on the first of these mechanisms, which does not rely on the existence of functioning complementary markets.

The simplest theoretical models of land tenure and investment attempt to capture tenure as an exogenous probability of losing the land, along with any fixed investments, before the profits of those investments can be fully realized. These models often make this assumption to instead focus attention on other features of the investment context (Jacoby and Minten, 2006), such as intra-household bargaining (Dillon and Voena, 2015). However, much qualitative evidence on customary tenure regimes in Sub-Saharan Africa has stressed the endogeneity of tenure security, which has been incorporated into quantitative models. Actions taken

as well as a desire to protect newly more valuable land from potential land grabs.

¹⁸They find that investment increases with length of occupancy, with a discontinuity at the 12 year mark when rights become recognized by law.

¹⁹In China, Li and Zhu (2023) for example find that a legal reform recognizing transfer rights increases investment, although land concentration seems to be an important mechanism, which may be less possible in Africa.

by the farmer, particularly certain investments in the land, demonstrate responsible use and defend against expropriation by others in the community who recognize these investments as land stewardship (Awanyo, 2009; Bambio and Bouayad Agha, 2018).²⁰ Deininger and Jin (2006) emphasize the visibility of an investment to others in the community as key to its efficacy in securing tenure rights, for tenure security is fundamentally about social recognition of rights. Place and Otsuka (2002) explore this concretely by contrasting three possible investments: planting trees, which is visible and thus reduces the probability of losing the land (a primary input in my model below); management effort, which is invisible and thus has no effect on tenure security; and fallowing, which in a tenure system predicated on land *use*, may actually increase the risk of expropriation.²¹

I now turn to my own theoretical model of tenure and investment, explicitly incorporating changing incentives for expropriation on land under different tenure systems. Consider a two-period model. In the first period, a farmer chooses to purchase and apply a short-term input, fertilizer, F_1 , and a long-term one, trees, T ,²² separately to a staple crop s and cash crop c planted on his or her exogenously given land L under tenure system h in order to produce according to $f_s(F^s, T^s)$ and $f_c(F^c, T^c)$.²³ A fraction $0 \leq \alpha \leq 1$ of the land is devoted to the staple crop, with the remainder $(1 - \alpha)$ used for the cash crop; this share is taken as given for the input use decisions.²⁴ The farmer also has some wealth endowment w . The farmer has a discount rate β for second period returns (the optimized expected second-period total profit is given by π_2^*). There is some probability that the farmer’s land will be expropriated before the second period, in which case fixed investments (trees) would also be lost. This probability is given by $(1 - \phi(T, h, vL))$ (where v is the per-unit value of the land) and may be a function of tree investment, tenure systems, and the value of the land. For convenience, I use the word ‘expropriate’ to capture both ‘horizontal’ expropriation (where others in the community take over the farmer’s use rights), as well as the exertion of sales rights by local elites without the permission of the land user (‘vertical’ expropriation). ϕ is written as a function of vL

²⁰Indeed, Zamchiya and Musa (2023) find in Zambia that a majority of investments undertaken were made in order to secure the land they were on.

²¹Goldstein and Udry (2008) focus on this latter effect in their work on tenure security in Ghana, where fallowing is the primary investment in land productivity. They focus on how an individual’s social status interacts with these incentives: “farmers who lack political power are not confident of maintaining their land rights over a long fallow. As a consequence, they fallow their land for much shorter durations than would be technically optimal, at the cost of a large proportion of their potential farm output.” This effect is driven by non-elite households on their land obtained through customary tenure. Social elites do not face the same disincentives to fallow, as they would be the ones to transfer unused land, so empirically they fallow land similarly regardless of how a parcel was acquired.

²²These inputs stand in for bundles of short and long-term investments, although trees may have particular tenure-enhancing effects not seen for other long-term inputs.

²³I leave this production function general, only assuming diminishing marginal returns to each input ($\frac{\partial^2 f}{\partial F^2} < 0$, $\frac{\partial^2 f}{\partial T^2} < 0$). These assumptions allow me to argue that in cases where the marginal productivity of a given input are lower, the level of that input used is higher, and therefore to predict how investment will respond to changing land values, which is the empirically relevant hypothesis, even without estimating a full agronomic production function which is beyond the scope of this model. The production functions for each crop may differ, and in particular may have different marginal productivities of inputs.

²⁴The model could be extended to a three period model, where α is chosen in the first stage prior to input use decisions, and retain its primary results.

as well as h , the tenure system, and T , tree investment. In particular, $\phi(T, h, vL)$ could have the following properties (laid out in more detail in appendix A):

$$\phi(T, h, vL) = \begin{cases} 1 & \text{if } h = 1 \text{ (Freehold),} \\ g(T, vL) & \text{if } h = 0 \text{ (Customary).} \end{cases}$$

where $\frac{\partial g}{\partial T} > 0$, as in Place and Otsuka (2002), and $\frac{\partial g}{\partial v} < 0$. Intuitively, this implies that freehold land is perfectly secure (and not responsive to either tree planting or land values). Customary tenure security, by contrast, is increasing in tree planting (which demonstrates responsible stewardship of the land and thus guards against horizontal expropriation by neighbors or family members), and decreasing in land values (which heighten incentives for elite expropriation for sale to outsiders).

After working through the main features of the model (a fuller, more formal treatment of the model, including assumptions and full solution, is presented in appendix A), I consider what different forms of this ϕ function could imply for the farmer's decisions as land pressures increase.

The first period optimization problem faced by the farmer is therefore

$$\begin{aligned} \max_{F_1^s, F_1^c, T^s, T^c} & \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T, h, vL) \pi_2^*(F_2^s, F_2^c, T^s, T^c) \\ \text{subject to: } & p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \end{aligned}$$

In the second period, the farmer only chooses fertilizer (F_2^s and F_2^c) and therefore maximizes according to the following:

$$\begin{aligned} \max_{F_2^s, F_2^c} & \alpha p_s f_s(F_2^s, T^s) + (1 - \alpha) p_c f_c(F_2^c, T^c) - p_F(F_2^s + F_2^c) \\ \text{subject to: } & p_F(F_2^s + F_2^c) \leq w_2 \\ & w_2 = w_1 + \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) \end{aligned}$$

Consider the case where the budget constraint does not bind: that is, the farmer can invest optimally in all inputs in both periods. The equilibria conditions are:

$$\begin{aligned}
\frac{\partial f_s}{\partial F_1^s} &= \frac{p_F}{\alpha p_s} \\
\frac{\partial f_c}{\partial F_1^c} &= \frac{p_F}{(1-\alpha)p_c} \\
\frac{\partial f_s}{\partial T^s} &= \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{\alpha p_s (1 + \beta \phi)} \\
\frac{\partial f_c}{\partial T^c} &= \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{(1-\alpha)p_c (1 + \beta \phi)}
\end{aligned}$$

Figure 1 depicts these equilibria conditions. Parameters γ and τ represent the total expected marginal productivity of inputs F and T , respectively; these marginal productivity curves differ for the staple s and cash crop c . The unconstrained farmer will choose inputs F and T for each crop such that the total expected marginal productivity is equal to the marginal cost of the input: at the intersection of the marginal productivity curve with the price of the input.²⁵ For fertilizer, depicted in the two left diagrams, optimal application per unit of land is the same between freehold land (above) and customary land (below); that is, $F^{s1} = F^{s0}$ and $F^{c1} = F^{c0}$.

A more formal solution can be found in appendix A.

3.1 Customary Tenure and Investment

The threat of future expropriation of the land is real for many farmers, particularly on customary land. This expropriation can happen with little to no advance notice, as it takes the form of the sale or physical expropriation of the land by someone other than the use rights holder, rather than the non-renewal of a contract.²⁶

The model captures this risk in two ways. Simple models of customary tenure and investment consider customary land to have a higher risk of alienation than freehold land: $\phi(T, vL|h = 0) < \phi(T, vL|h = 1) \forall T, v, L$, with $\frac{\partial \phi}{\partial T} = 0$. That is, for any level of investment in trees or land values, the risk of expropriation is higher on customary land, and this risk does not change in response to investment in the land. This decreases the expected total marginal returns to investment in trees, as shown by a shift towards the origin from τ to τ'' , decreasing the optimal investment in trees. Considering the equilibria conditions, the risk of expropriation $\phi < 1$ deflates the denominator of $\frac{\partial f}{\partial T}$, meaning the marginal productivity is higher, which

²⁵If output prices are low enough, or if input prices are high enough, optimal application may be zero, particularly for lumpy investments such as trees. However, the impact of changing land pressures that we will examine should still hold, even if farmers do not cross the threshold to non-zero investment.

²⁶As Abera and Chemin (2021) note, the power differentials inherent in elite expropriation also mean that expropriators may not face any sanctions for their behavior. A useful analogy is that of the roving bandit (Olson, 1993): local elites had been functioning as stationary bandits, in an ongoing relationship with the primary use rights holder. The new temptation of sales to outsiders, however, may shift their role to that of a roving bandit who expropriates.

occurs at lower levels of investment in trees given diminishing marginal returns.

Drawing upon qualitative understandings of customary tenure, though, more subtle models have sought to capture the endogenous nature of tenure security on customary land. Place and Otsuka (2002), for instance, model the probability of losing land as a decreasing function of planting trees, a visible investment in the land. In my model, this implies that $\frac{\partial \phi}{\partial T} > 0$, which would attenuate the leftward shift of the τ curve²⁷ (and could perhaps even lead to an ‘overinvestment’ in trees relative to the unconstrained optimum if the tenure-enhancing effect is large, although empirically this is rarely documented). This combined effect leads to the marginal returns to trees indicated by τ' .

The equilibrium determined by the marginal productivity of trees under (less-secure) customary tenure, then, likely sees under-investment in trees relative to the allocation under freehold tenure ($T^{s0} < T^{s1}$ and $T^{c0} < T^{c1}$). I will empirically test this using data from Uganda.

3.2 Expropriation Risk and Changing Land Pressures

Customary tenure is not simply insecure tenure. Above, I showed that an attention to the complexities of how customary tenure functions can have important implications: visible investment decisions such as planting trees can demonstrate responsible land use (a condition of tenure), thus endogenously strengthening tenure rights. Recall that by planting trees on a plot, a farmer demonstrates to the community that it is under use. If another member of the community tries to encroach on the plot, these visible investment decisions can be used as evidence of responsible land use, and thus make it more likely that disputes within the community will be resolved in the farmer’s favor.

The model so far has ignored the particular bundles of rights and their distribution among individuals that are hallmarks of customary tenure systems. That is, many customary tenure systems in Sub-Saharan Africa, and those in Uganda in particular, do not vest all types of rights over a given piece of land in the same individual or household. Primary use rights may be held by one household, while other members of the lineage or extended family have access rights or even claims to certain plants (Howard and Nabanoga, 2007). Importantly for this analysis, local elites (such as lineage heads or traditional chiefs) often hold transfer rights over large areas of customary land.

These decentralized rights do not in themselves cause major tenure insecurity (Bruce and Migot-Adholla, 1994); indeed, they can provide much-needed flexibility to adjust to changing populations. Mechanisms, such as planting trees, have developed to allow use-holders to demonstrate tenure against other claimants, with local elites serving as arbiters and rights-holders for the community.

²⁷Equivalently, it decreases the numerator of $\frac{\partial f}{\partial T}$, meaning the marginal productivity of trees are lower, occurring at higher levels of tree investment – attenuating the effect of ϕ on the denominator discussed above

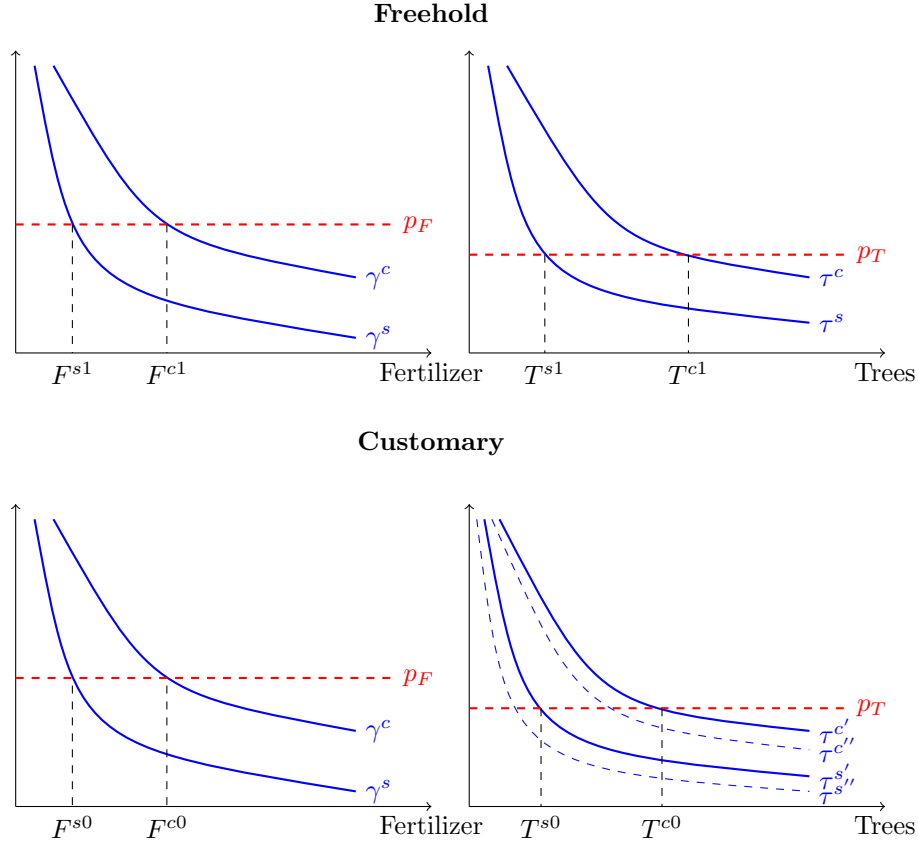


Figure 1: Equilibria conditions for fertilizer (left) and trees (right) investment on freehold (top) vs customary (bottom) land, for staple crops s and cash crops c . Expected total marginal productivity of fertilizer depicted by γ , and of trees depicted with τ . Optimal investment is where each curve crosses the dashed line, representing the price of the input. Optimal allocation of fertilizer is the same across tenure types, but on customary parcels, the risk of expropriation before the second period could lower the marginal productivity of trees to τ'' , implying lower investment in trees on customary parcels relative to freehold. If trees have a security-enhancing effect on customary parcels, this shift towards the axis is attenuated, represented by τ' .

However, there has been recent concern in policy spheres about the potential for local elites to make deals with outsiders (often national or international investors) that alienate local rights-holders. While the details of such deals, termed ‘Large-Scale Land Acquisitions’ (Smalley and Corbera, 2012; Purdon, 2014), are beyond the scope of this model, they represent but one example of local elites using their traditional rights to the land for their own gain.²⁸ Outside investors often find it difficult to navigate the complex realities of customary rights, and thus they may not realize that they are expropriating land from existing users without their consent.

In the context of this model, increasing land pressures has two main impacts. First, the relative output price of cash crops increases with greater access to urban markets.²⁹ Secondly, the increased access to urban markets also brings in outsiders who may seek to acquire land locally. These outsiders may find it difficult to determine who owns undocumented customary land. Local elites may not have found it profitable to exercise their traditional rights in an inactive land market, preferring instead to maintain the traditional status quo. However, as land values rise (whether due to offers from outside investors, population pressures from growing urban areas, or other exogenous forces that introduce potential outside buyers who are less familiar with the customary) they may be more likely to expropriate from the land user in order to alienate the land to others.³⁰

The land user (farmer/decision-maker) can anticipate these changing incentives for their co-rights-holders, as captured in $\frac{\partial \phi}{\partial v} < 0$. Rising land values may mean that no matter the level of their own investment in production, the risk of expropriation increases.³¹ Figure 2 illustrates this: the higher risk of expropriation could outweigh increased output price impact of increasing land values, and perhaps even on net decrease longer-term investments in customary land. The shift from T^{c1} to $T^{c1'}$ and F^{c1} to $F^{c1'}$ reflects the effect of increasing land values on investment on freehold land: investment in both fertilizer and trees increases, due to higher prices for cash crops. For customary land in Figure 2, increasing the value of the land does not simply shift the marginal productivity of trees on cash crops rightward, but changes the shape due to the

²⁸It could be argued that this is in fact a reinterpretation of their customary rights, shifting from custodian of the land to ‘owner’ in a more Western, exclusive, sense.

²⁹Although greater access to markets may also increase the price of the staple crop, and perhaps decrease input prices of fertilizer and trees, in relative terms the largest effect is the increase in the price of the cash crop. On the other hand, greater access to urban markets may increase the opportunity cost of on-farm labor (but also may increase the supply of hired labor), which could be thought of as an change in the shadow price of labor-intensive inputs such as trees. This effect should be common for all parcels in an area, though, and therefore should not differentially impact customary vs. freehold parcels within the same geography, the comparison I am making.

³⁰An alternate, complementary, understanding of this process (explored in Honig (2022)) is that in isolated communities, returns to elite status are primarily social – respect and prestige from their community. But as outsiders penetrate the community, that social prestige may be less valuable or obtainable from a community in flux. This may make the monetary benefits of expropriation seem even more valuable.

³¹Indeed, these rising land pressures may also weaken the relationship between tree planting and tenure security, $\frac{\partial \phi}{\partial T}$, as community norms around responsible land stewardship may not outweigh individual expropriation incentives of more valuable land. This would increase the numerator of the equilibria conditions for tree planting, and therefore mean yet lower optimal investment in trees on customary land.

effect of $\frac{\partial \phi}{\partial v}$ discussed above. Looking at the equilibria conditions, tree planting on staple crops on customary land will only be impacted by the decrease in ϕ , increasing the marginal productivity of trees (implying a decrease in the level of trees planted); for cash crops on customary land, this could be offset somewhat by the increase in p_c . In sum, though, we would expect a smaller increase in tree planting on customary land due to an increase in land pressures than on freehold land.

4 Data

To test this model empirically, I use data from the baseline survey I conducted for a proposed impact evaluation of Uganda’s Agriculture Cluster Development Project (ACDP).³² This baseline survey was conducted in early 2019 in four districts throughout Uganda. Respondents were chosen in a two-stage process: first, we randomly selected 133 registered farmer organizations that had at least 20 members in the target districts, and membership rosters were drawn up. Member households were then randomly selected to be interviewed, with a total of 2,189 interviews in the baseline survey used here.

The districts selected for the impact evaluation are located throughout the country and thus encompass some of the diversity of customary tenure arrangements in Uganda. Selected districts are located in the east, southwest, center (where *mailo* land predominates), and north (where instability has led to many internally displaced people).

As part of the impact evaluation, the Ministry of Agriculture, Animal Industry and Fisheries and a team from UC Davis conducted an extensive baseline survey,³³ which has data well-suited to the structure of this model. Given that the program was designed to induce investment in agriculture, the survey has an exhaustive plot-level elicitation of all inputs and outputs as well as continuous measures of the value of investment. Furthermore, the survey asks detailed questions about the tenure status of each agricultural parcel farmed by the respondent household. These questions ask about the tenure system the parcel is under; which individual members of the household are owners; what kind(s) of documentation exist for the parcel, if any, and whose names are on the documentation; who within or outside the household holds particular rights

³²This project, supported by the Government of Uganda and the World Bank, aims to provide farmers in five crop-specific ‘clusters’ (maize, rice, beans, cassava, and coffee) with the resources needed to move from subsistence farmers to commercialized producers. To that end, the project provides subsidized farm inputs to selected farmers, improves agricultural infrastructure, and supports post-harvest handling technologies. The impact evaluation was planned as part of the pilot phase of the project, focusing on the provision of electronic vouchers for subsidized inputs that are redeemable at certified local agro-input dealers. The impact evaluation was planned in 4 districts throughout Uganda, each assigned one of 4 target crops (maize, rice, beans, and coffee; cassava was omitted as it has a longer growing season). The impact evaluation randomly assigned farmer organizations eligible for the ACDP subsidy to one of two treatment groups (defined by the timing of benefits) or to a ‘downstream’ control group. Approximately 36 farmer organizations were assigned to each treatment arm, from each of which an average of 20 members were selected to be surveyed throughout the four districts.

³³Subsequent mid-line and end-line surveys were planned but were cancelled along with the implementation of the Impact Evaluation in November 2020, due to compliance issues and COVID-19. The baseline data I use was entirely collected pre-intervention, so there should be no impacts of the intervention on input use.

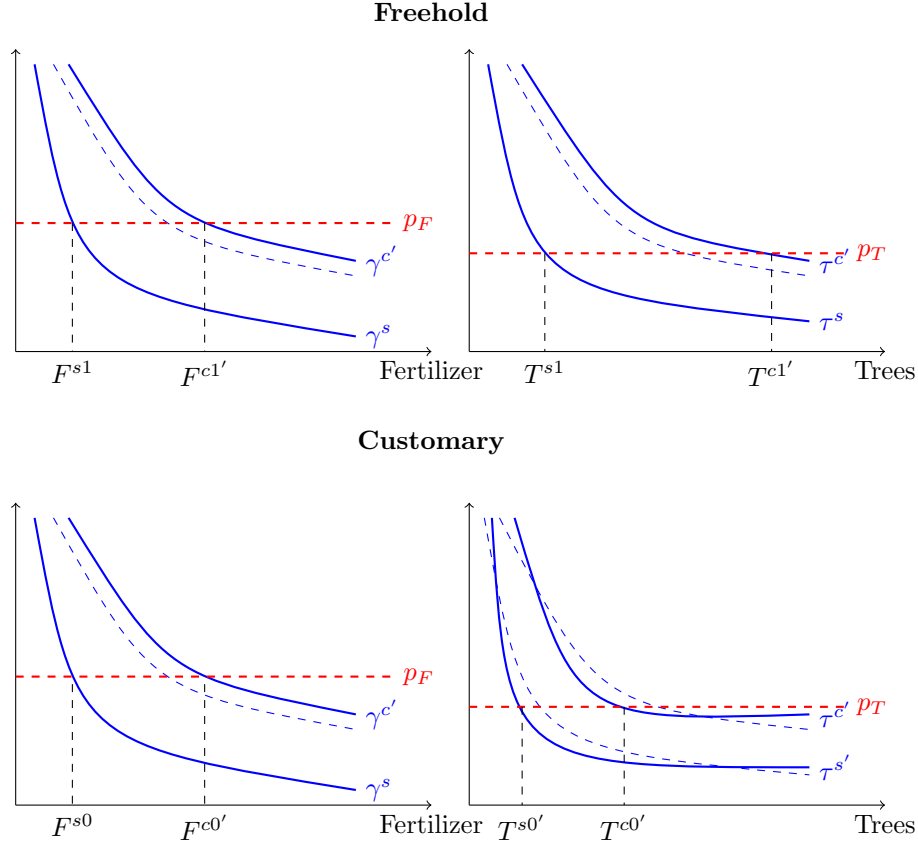


Figure 2: Equilibria conditions for fertilizer (left) and tree (right) investment on freehold (top) vs customary (bottom) land, for staple crops s and cash crops c , in a high land pressure environment. Expected total marginal productivity of fertilizer depicted by γ , and of trees depicted by τ . Optimal investment is where each curve crosses the dashed line, representing the price of the input. As land pressures increase, the relative output price of cash crops increases, raising the expected total marginal productivity of inputs on the cash crop from the dashed line to $\gamma^{c'}$ or $\tau^{c'}$, and thereby increasing optimal fertilizer and tree application to the cash crop on freehold land and fertilizer application to the cash crop on customary land. On customary land, however, increasing land pressures also increase the likelihood of expropriation, decreasing the expected total marginal productivity of trees on customary land. This results in the changed shape of the marginal productivity curves in the bottom right subfigure, with a lower optimal allocation of trees to both crops on customary land. This means that although optimal fertilizer use on customary parcels increases to $F^{c0'}$, optimal tree planting does not increase with higher land values and is instead depicted by $T^{s0'}$ and $T^{c0'}$.

(such as transfer rights); and questions on perceived tenure security. Finally, the survey asks about land rental prices in the region and contains GPS coordinates for the household, allowing me to match households to other geospatial datasets. For most specifications, I use a dataset of 3,076 parcels; most were observed over two seasons, for a total of 5,760 parcel-season observations which I pool.

To some extent, this sample is positively selected in comparison to all farmers in Uganda: they are members of farmer groups, indicating a higher likelihood of participating in output markets and perhaps higher farming skill, although farmer groups are relatively common in Uganda.³⁴ This positive selection can be seen in rates of input use in the ACDP data when compared with the nationally-representative Living Standards Measurement Survey (LSMS) conducted by the Uganda Bureau of Statistics and the World Bank: only 4.6% of parcels have organic fertilizer applied and 2.6% inorganic fertilizer in the 2019 LSMS, as compared with 11.5% and 10% of parcels in the ACDP data, respectively. This higher rate of input use provides a key advantage over the more widely used LSMS data: I can estimate the interacted models that emerge from my model due to the variation in the outcome. However, it comes at a potential cost of external validity.

I use two measures of the external pressure on land values modeled above. First, I use the inverse travel time to Kampala, the national capital, adapted from Muller-Crepon et al. (2021) and Muller-Crepon (2021). This measure is constructed using historical road maps from Michelin, which Muller-Crepon et al. (2021) digitized and created a road atlas for all of Africa from 1966 to 2015, akin to a historical Google Maps. In Muller-Crepon (2021), he then estimates the shortest travel time via roads of different qualities to the national capital for each 5x5 km grid cell.³⁵ Both Muller-Crepon (2021) and Muller-Crepon et al. (2021) use the inverse of travel time to the capital as proxy of state reach in a local geography, but it also represents the degree to which outsiders such as urban residents of the capital may be interested in purchasing local land: precisely the concern of customary landowners that creates insecurity. Figure 3 plots both the average travel time to Kampala, as well as the 10 year decrease in that time, for each parish of the country, with ACDP districts highlighted in black (maps of each measure individually can be found in figures 8 and 9).³⁶

I match households to a grid cell using GPS locations collected during the survey (taken at the household residence, not at agricultural fields which could be at some remove). Following Muller-Crepon et al. (2021), I then compute the ‘access’ to the national capital in 2015 for each household as $(1 + time_j)^{-1}$.³⁷ The

³⁴In the 2019 Living Standards Measurement Survey, 28% of communities report having a farmers’ group; 43% report a savings & loan group that was identified by subcounty extension workers as serving as a farmers’ group for the purposes of ACDP targeting.

³⁵This is approximately 2.5 arc-minutes at the equator, which Uganda spans.

³⁶I focus on travel time to Kampala rather than to regional capitals in this analysis, as I want to isolate away from the politically-driven administrative unit proliferation that has occurred in Uganda in the recent past. Although a new district capital may be identified close to respondents, these new centers do not yet have the population to support substantial outsider demand for land.

³⁷This represents the most recent year for which Muller-Crepon (2021) has data on travel time to national capitals, regional

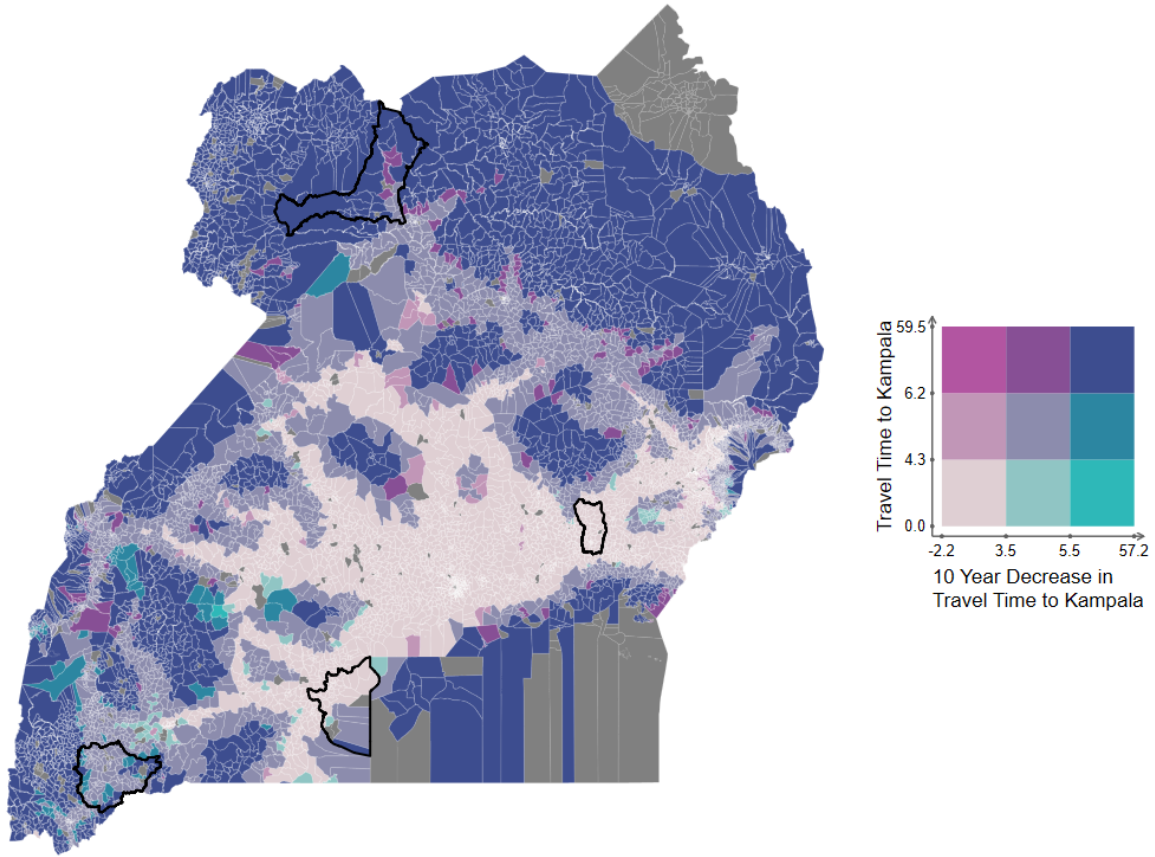


Figure 3: Map of the average travel time in 2015 to Kampala, as well as the 10-year decrease in that travel time, for each Parish in Uganda, with ACDP districts highlighted in black. Administrative units in Uganda extend into Lake Victoria to the national border, to include several small islands.

correlation between per-acre shilling rental prices and access to Kampala is low (.08) but increases to .21 when I winsorize several outliers in estimated rental prices (among those who own their land and are estimating how much they could rent it out for)³⁸ and to a very high .93 when I consider district median rental prices, suggesting that it captures broader pressures than individual prices which may be influenced by soil quality or other agricultural value considerations.

The second measure I use attempts to isolate changing pressures on local land, which is a more direct representation of the pressure on local elites to expropriate and sell customary land to outsiders. I use the historical (10-year) decrease in travel time to Kampala, due to the construction or upgrading of roads, relative to the 2005 travel time to Kampala. This measure, also constructed using historical spatial data from Muller-Crepon et al. (2021), captures the extent to which elites face *new* opportunities to expropriate, which should have a strong impact on perceptions of tenure insecurity. I match this measure spatially in the same way as the inverse travel time variable described above.

5 Empirical Strategy

To test the predictions outlined in the model, I examine differential patterns of short- and long-term agricultural investments on parcels under different tenure regimes as land pressures exogenously increase. These predictions go beyond existing models of land tenure and investment by considering how overlapping rights-holders respond to rising land prices, and thus how incentives may change differently on customary land. Commonly-used empirical strategies model tenure *security* (the expected probability of expropriation before realizing the fruits of investment) but then only empirically measure rough proxies such as tenure *type*. Improving over this, I explicitly model the relationship between tenure *type* and investment (operating through tenure security). The theoretical model explored above shows how a rough proxy may be insufficient: in low land-value environments, customary land may be equally secure as freehold, but as land values increase, customary land may become less secure.³⁹

Outside this advancement, I use empirical methods generally accepted in the literature. At heart, the empirical specification involves regressing a given investment measure on a continuous measure of land values interacted with a dummy for a parcel being under customary (as opposed to freehold) tenure. Investments should rise with land values on freehold land due to higher output prices for cash crops (with a higher

capitals, and distance to markets, and thus the closest to the ACDP baseline survey in 2019.

³⁸In most empirical specifications using rental prices, I standardize them to deal with outliers rather than using a more ad-hoc method such as winsorizing. However, for this correlational check with shilling prices, I assign all per-acre rental prices below 10,000 shillings (85 parcels) a value of 10,000 shillings per acre (lower than the 5th percentile), and the top 1.6 percent of per-acre rental prices the 98.4th percentile value.

³⁹The lack of results in previous studies which elide tenure security and tenure type may be understood, then, as an issue of this poor proxy.

productivity of inputs); on customary land, however, this effect is attenuated by the escalating insecurity caused by local elites' incentives to expropriate newly-valuable land to sell to outsiders. The coefficient on the interacted term of customary tenure with a measure of land pressure is of primary interest: is the response to rising land pressures statistically different on customary parcels than on freehold ones?

Several concerns are apparent with this strategy. First, the tenure regime of particular parcels may not be exogenous (for example, households may be more likely to pursue titling for higher-quality plots). Deininger and Ali (2008) argue that Uganda has what amounts to an "exogenous historical assignment of land rights... and the absence of readily available opportunities to change the tenure status of occupied land to full ownership imply that the case at hand can be considered akin to a natural experiment."⁴⁰ In appendix C.7.1, I explore differences in parcel characteristics by tenure type, and although there are some observable differences in table A33, the results are robust to controlling for these characteristics in tables A34 - A38. Second, there may be unobserved heterogeneity in household farming decisions caused by factors such as household shadow prices or farming skill, which may be correlated with the tenure of parcels farmed by households. Ideally, this would be addressed using a household fixed-effects strategy. Unfortunately, of the 2,189 households surveyed, only 6.5% of respondents reported having parcels under both freehold and customary tenure, which is much lower than the rates observed in the LSMS-ISA in Uganda, a high-quality nationally-representative dataset (Deininger and Ali, 2008). Furthermore, these households are concentrated in two of the four surveyed districts. This limits the power of a household fixed-effects model to detect changing investment patterns as land prices increase.

Despite this, the ACDP data sampling strategy involving farmer groups allows me to extend the intuition of household fixed effects to farmer group fixed effects. These farmer groups are co-located, self-selected groups of farmers who can purchase inputs in bulk, share farming methods, and aggregate outputs in order to receive better output prices, in addition to meeting regularly for mutual support. Therefore, it seems plausible that many unobservable characteristics are shared by members of the same farmer group. By including a farmer group fixed effect, I am able to control for a portion of the unobserved heterogeneity while still retaining some identifying variation. However, this does restrict the identifying variation almost to zero for regressions using the inverse travel time. This is measured at a fairly coarse geographic scale, meaning that most farmer groups are all within the same grid square or parish. Including farmer group fixed effects when using these geographically coarse measures of land pressures would mean I am identifying off of within-group noise from the few groups that have members in grid cells with different travel times, which could give misleading estimates. Therefore, my preferred specification includes farmer group fixed effects for

⁴⁰Perego (2019) describes the costly and onerous process required to register customary land, which provides substantial barriers to titling.

regressions using the standardized rental price as the measure of land pressures (column (2) in results tables 1 - 6) but without the fixed effects for regressions using the inverse travel time or changes in travel time (columns (3) - (4) in tables 1 - 6). In interpreting coefficients on rental price with the farmer group fixed effects, I identify off of within-group differences in land pressures; for other measures, the results are more between-groups or regions. All specifications cluster standard errors at the larger parish level, to account for geographic relationships between households. In appendix C.5, I present results with a more traditional parish fixed effects specification with land values (table A19) for robustness, as well as computing spatially dependent standard errors as in Conley (1999) (tables A5 - A9 in appendix C.2).

The primary results that I present have land pressures entering linearly, with a constant marginal pressure on either customary or freehold land. This may not be the appropriate specification: it may be that elite incentives to expropriate only begin at intermediate or relatively high land pressures, or alternately that in high land pressure environments, the likelihood of expropriation of customary land has been ‘priced in’ to investment behavior, meaning the marginal effect of increasing pressure is minimal. In order to explore this, I examine whether the relationship differs on either side of a travel time threshold identified in interviews as being relevant for external land demand from Kampala, 2 hours travel time. I separately estimate the responsiveness of investment to land pressures on customary and freehold land on either side of the threshold.

5.1 Hypotheses

First, I illustrate the hypothesized comparative statics from the model represented in figures 1 and 2 and identify which terms would capture them in regressions. Consider the following cross-sectional empirical specification which closely follows the model discussed above:

$$I_i^Q = \gamma_0 + \gamma_1^Q D_i + \gamma_2^Q \bar{v} + \gamma_3^Q D_i * \bar{v} + \beta' X_j + \eta_f + \epsilon_{fi} \quad (1)$$

where I_i^Q is a measure of investment of type Q (either short- or long-term) by household j (a member of farmer group f) on parcel i ; D_i is a dummy equal to one if the parcel is customary; $\bar{v}$ is the exogenous component of land price in the locality; X_j is a vector of household characteristics (a probability of poverty index, total household landholdings, and dummies for how the parcel was acquired); and η_f controls for farmer-group fixed effects. This equation would be separately estimated for long and short-term inputs.

The particular hypotheses predicted by the model are derived in the mathematical appendix. Here I summarize proposed tests of the predicted signs of the following in cross-sectional data:

1. $\left. \frac{\partial F}{\partial v} \right|_{h=1}$: That is, how does short-term input use vary with respect to land values on freehold land?

The model would predict that *ceteris paribus*, farmers would use more short-term inputs as land values

increase. This is tested by examining if $\gamma_2^S > 0$.

2. $\left. \frac{\partial T}{\partial v} \right|_{h=1}$: That is, how does long-term input use vary with respect to land values on freehold land?

The model would predict that *ceteris paribus*, farmers would use more long-term inputs as land values increase. This is tested by examining if $\gamma_2^L > 0$.

3. $\left. \frac{\partial F}{\partial v} \right|_{h=0}$: That is, how does short-term input use vary with respect to land values on customary land?

The model would predict that *ceteris paribus*, farmers would use more short-term inputs as land values increase. This is tested by examining if $\gamma_2^S + \gamma_3^S > 0$. The extent to which short-term input application responds more strongly to land pressures on freehold vs customary land is given by $\gamma_3^S > 0$

4. $\left. \frac{\partial T}{\partial v} \right|_{h=0}$: That is, how does long-term input use vary with respect to land values on customary land?

The model would predict that *ceteris paribus*, farmers would use fewer long-term inputs as land values increase. This is tested by examining if $\gamma_2^L + \gamma_3^L < 0$.

with the last of these being the primary innovation of this model. Without considering the changing incentives of local elites, the impact of rising land values on customary land would parallel that on freehold. This stems from the qualitative fact that tree planting behavior is effective at securing tenure within the community, but less so to outside investors who find this demonstration of rights illegible. Importantly, the model would predict that γ_3^L , which captures the difference in responses to land pressures between freehold and customary land, would be negative, whereas γ_3^S is likely zero or positive.

5.2 Measuring Tenure Security

In taking stock of the body of empirical evidence on tenure security and investment, several papers have noted the variety of definitions of tenure security as well as proxies used empirically (Arnot et al., 2011; Deininger and Ali, 2008). Common empirical measures include the existence of legal title, duration of tenure, method of acquisition, tenure type, existence of a conflict over the land, subjective perceptions of tenure security, and existence of particular rights (often transfer rights) (Arnot et al., 2011). It is often left implicit how these proxies are related to tenure *security*, which is the parameter modeled. Fenske (2011) notes that the choice of proxy does seem to be related to the effect found, and therefore this is a crucial choice. I do have a measure of subjective tenure security, which does respond to changing land pressures as in the model; in appendix C.6.1, I also show in table A20 that alternate proxies such as documentation and past experiences of disputes have the expected correlations with perceived tenure security.

My model addresses concerns with measures of tenure security by explicitly considering the relationship of the empirically observable tenure type to the fundamental parameter of interest, tenure security. The

importance of this consideration is illustrated by the final hypothesis, that customary tenure exists in an institutional context that responds differently to changes in land value, and therefore the relationship between tenure type and security is different in high and low land value areas. Arnot et al. (2011) distinguish between the ‘content’ and ‘assurance’ aspects of tenure, and rather than eliding between those, I consider how the two are related.

5.3 Measuring Investment

Another point at which many papers allow their model and empirics to diverge is in the measurement of agricultural investment, the outcome of interest. Because many smallholders operate low-input, often subsistence farms, it is difficult to measure investment well. Therefore, many papers have used binary investment outcomes (Fenske, 2011). I present binary outcomes using a linear probability model in order to allow for farmer group fixed effects (much as Deininger and Ali (2008) do).

However, the ACDP data also allows for continuous measures of investment, although these were measured with some error. Fenske (2011) has argued that measuring investment in a continuous way can make a substantial difference to the results. In particular, it may attenuate coefficients of interest: in my model, although changing land pressures may increase optimal investment, the changes may not be large enough to lead a non-investing farmer to positive investment, particularly in a lumpy technology such as trees, which would be observationally equivalent to no impact. Fenske (2011) argues for the trimmed least absolute deviations estimator to deal with the prevalence of zeroes in types of investment. This is the panel equivalent of the Tobit estimator (Honore, 1992); as I only have a cross-section of data, I use the Tobit for continuous investment measures; results are in appendix B. Coefficients on a Tobit model are interpreted as in a linear model; however, the linear effect is on the uncensored (non-zero) outcome. Broadly, these results are consistent with those in the binary outcomes case.

5.4 Land Values

Importantly, the model presented above takes land values to be an exogenous force shaping the incentives faced by farmers. This modeling choice was driven by the qualitative literature, which has focused on the breakdown of customary institutions caused by outside pressures, such as the rapid urban expansion in much of Africa or international large-scale land acquisitions. In reality, the value of an individual parcel of land is a product of its particular features, including any investments made and tenure type.⁴¹ In my results, I

⁴¹The ACDP survey also asks for self-reported rental values, including estimated values for parcels not actually rented, which could be influenced by farmer awareness of land market conditions (and their human capital).

present some specifications using standardized per-acre land rental values⁴² (in appendix C.6.2, I also use alternate measures of land prices, including predicted prices for all parcels, median village prices, and non-standardized rental value). These results carry the crucial caveat that land prices are likely a result of land quality, which may change optimal investment behavior through pathways other than those modeled.

This is why I also present results where I proxy for the exogenous pressures on land values with the inverse travel time to the national capital or the relative 10-year change in travel times to the capital. These measures should impact agricultural investment through the pathways described in the model: certainly, more (peri-)urban areas have more active credit markets, which would relax the credit constraint faced by the household, and a pool of urban buyers is the canonical situation incentivizing local elites to expropriate customary land. The process of urbanization, though, should not impact agricultural investment through other channels, such as via an increase in total factor productivity. These geospatial measures of land pressures should not be endogenous to the tenure or other characteristics of a particular parcel, and therefore should not suffer from the same endogeneity bias.

These measures could potentially be used as an instrument for local land values (as seen above, they are correlated with regional patterns in land values and would therefore be valid); however, I choose to treat them as proxies for the underlying phenomenon of interest, external pressures and demand for land. Rental values for individual parcels are perhaps an even more problematic proxy for the forces modeled, so I focus on these more exogenous measures which should be better-identified. Nevertheless, these results may not be strictly causal; however, they can be understood as descriptive of patterns consistent with my theoretical model. I have constructed all measures of land pressures such that higher values represent greater pressure on land: higher standardized rental prices, larger inverse travel times (equivalent to shorter travel times to Kampala), or greater decreases in travel time over the past 10 years relative to 2005 times.

6 Results

In this section, I present results for a variety of agricultural investments, following the empirical strategy in equation (1). The model predicts that long-term inputs respond to rising land values more strongly on freehold land than on customary land due to the attenuating effect of increased insecurity on customary land induced by elite incentives to expropriate.

Before turning to the regressions with individual agricultural inputs, consider figures 4 - 6. Each of these presents the difference in the coefficients for (binary) input use between freehold and customary parcels as a function of land values in the area (measured by rental values in figure 4, the inverse travel time to Kampala

⁴²I standardize rental prices by first assigning a parcel the median per-acre rental price in its district if it is missing a rental price, then subtracting the full sample mean per-acre rental price and dividing by the full sample standard deviation.

in 5, and the 10-year relative decrease in travel time in 6).⁴³ For short-term inputs such as pesticides and inorganic fertilizer, the model would predict equivalent responses to rising land pressures on all parcels, which would appear as a flat line on these graphs. As can be seen, however, there is a slight divergence in short-term input use on freehold vs. customary parcels as rental prices or the accessibility to Kampala rise, reflected by a positive (although shallow) slope in these figures for short-term inputs. This result, explored further below, can be explained by a shift toward input-intensive cash crops (both short- and long-term) near urban markets.⁴⁴ As urban markets expand, so does the local demand for cash crops (such as vegetables and fruits which are difficult to transport long distances). Farmers facing this increased demand therefore shift their production away from staples and towards cash crops. This shift is stronger on freehold parcels, perhaps because such parcels remain secure. Cash crops, in turn, may have a higher return to inputs, and particularly to pesticides, than staple crops (Riwthong et al., 2017). This shift towards cash crops with increasing land pressures is not captured in my model, where the share of land devoted to cash crops, α , was fixed for simplicity.

If the model holds, this effect of changing crop choices on long-term inputs should be compounded by the direct impact of additional insecurity on customary land prompted by rising land pressures. In figures 4 - 6, there is a stronger response for trees, the most long-term of the inputs measured, than for short-term inputs. Visually, the insecurity effect can be seen in the difference in slopes between the short- and long-term inputs in these figures.

Figures 4 - 6 also depict the difference in reported insecurity between freehold and customary land. This measure is discussed in further detail below, but note that at low land pressures, parcels under customary and freehold tenure are perceived as roughly equivalently secure. As pressures rise, respondents report higher perceived likelihoods of losing their customary parcels without an equivalent rise on freehold parcels.

In each table below, I present several different specifications; additional robustness checks are provided in appendix C.⁴⁵ I include three distinct measures of land pressures. First, I use the standardized per-acre

⁴³I separately regressed input use on the measure of land pressures for customary and freehold parcels, and then plot the difference in coefficients from each of these regressions. This is in contrast to the regression tables below, where I include all parcels and use interactions to model the relationship between land pressures, tenure type, and input use.

⁴⁴An alternate explanation could be different collateralizability of customary and freehold parcels. Farmers may be less able to internalize rising land values to relax their liquidity constraint on customary parcels. This would be less concerning if farmers held parcels under different tenure systems, due to the fungibility of loans. When I examine household credit usage in my sample, those in areas closer to Kampala are actually slightly less likely to take out loans; the relationship between credit usage and normalized individual rental prices is insignificant. This suggests that collateral is not driving the results for short-term inputs. Yet another, empirically indistinguishable, mechanism would be sectoral misallocation as a result of land and labor market restrictions (Adamopoulos et al., 2022), which could act more strongly in more remote areas; in my data, however, hired labor usage is near-universal, suggesting labor market restrictions do not bind.

⁴⁵These include exploring tenure security's relationship to land pressures across an expanded geographic sample that was part of a related survey, using Conley (1999)'s spatially dependent standard errors, splitting the sample with binary measures of land pressures rather than continuous measures, and including parish fixed effects. In Appendix C.6, I explore alternate measures of land values either derived from the survey, looking at regional market access, or using a predicted probability of urbanization from Seto et al. (2012). In C.7, I include various additional controls, such as parcel characteristics, excluding the possibly endogenous control of whether a cash crop was grown on the parcel, and controlling for reported individual access to markets: the reported one-way travel cost to either the nearest agroinput dealer or the nearest village in which crops can be

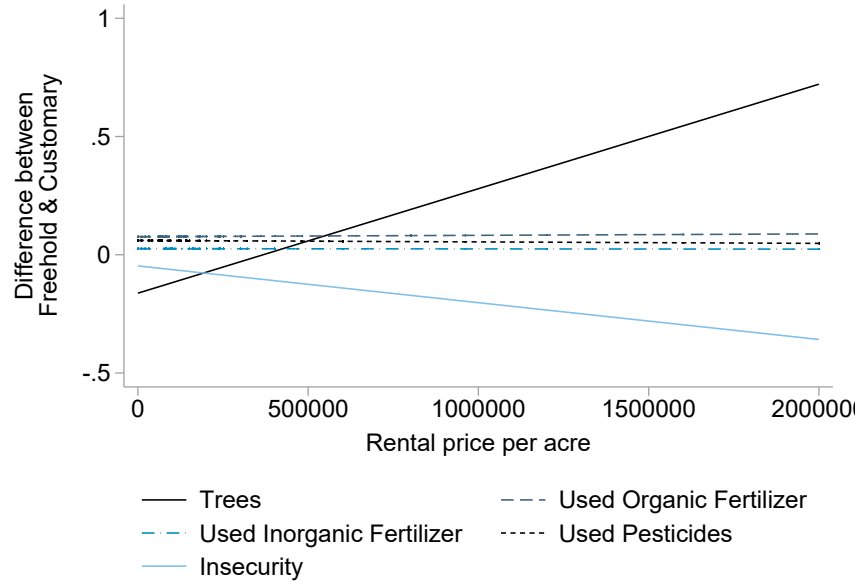


Figure 4: Disparities in tenure types increase with rental values more strongly for long-term than short-term inputs. Customary parcels become relatively more insecure than freehold as rental prices increase.

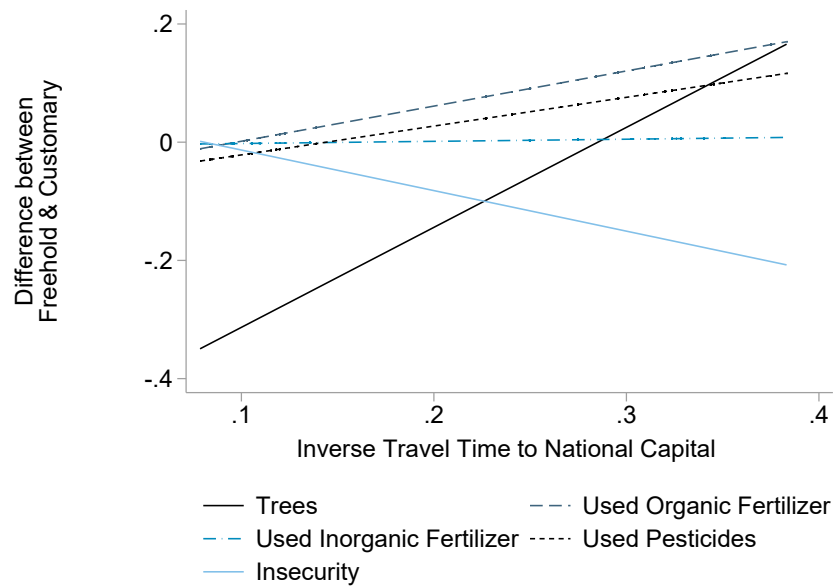


Figure 5: Disparities in tenure types increase with the inverse of travel time to Kampala (and thus demand for land by outsiders who may not understand local customary tenure) more strongly for long-term than short-term inputs.

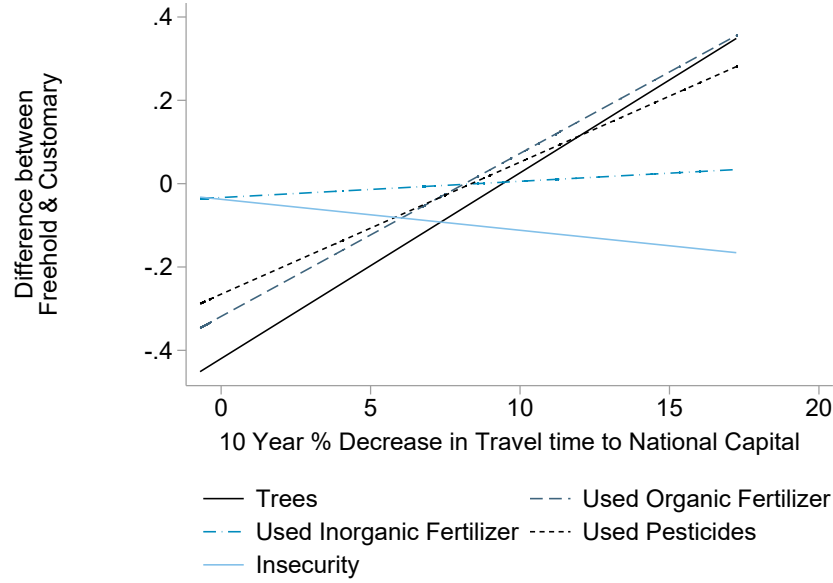


Figure 6: Disparities in tenure types increase with a greater percentage decrease in travel time to Kampala, a measure of increasing demand for land by outsiders, more strongly for longer-term inputs. However, there is a relatively large difference in responses for pesticides.

rental price in shillings as reported for a given parcel (either actual paid rental price or estimated potential rent if the parcel were rented out, and predicted using similar nearby parcels if the respondent did not report a rental price). As discussed above, this measure is not solely driven by exogenous land market pressures, and it could be capturing unobserved land quality (or even influenced by prior long-term investments in the land).⁴⁶ Therefore, I also show results using the inverse travel time to the national capital and 10-year decreases in travel time to Kampala (proportional to 2005 travel time) as proxies for underlying pressures, particularly outsider demand for land. These measures of land values are interacted with a dummy for the tenure status of a parcel, so the coefficient on ‘Land Value’ represents the responsiveness to land values on freehold land, while ‘Customary*Land Value’ represents the difference in that responsiveness for customary parcels.⁴⁷

In all regressions shown, I control for household characteristics: the Probability of Poverty Index (PPI) as well as for a household’s total landholdings in acres. The PPI uses low-cost survey indicators to estimate the likelihood that a household has consumption below a given poverty line, and it can be interpreted ap-

sold. These controls do not meaningfully change the results of the paper, although they do result in a substantial loss of sample size, hence their relegation to the appendix.

⁴⁶In appendix C.6.2, I show that results are similar when using predicted rental prices for all parcels, predicted using the probability of urbanization, latitude and longitude of the household, tenure status, existence of trees on the parcel, and village fixed effects.

⁴⁷Although conceptually similar to the slopes in figures 4 - 5, they are computed differently, include the addition of controls, and use clustered standard errors.

proximately as a wealth index (with higher values indicating a lower likelihood of poverty) (Schreiner, 2012). Encouragingly, the coefficient on PPI accords with intuition: wealthier farmers are generally more likely to use agricultural inputs, with the exception of trees, which may be more important for their security-enhancing functions for poorer households. I also control for how the parcel was acquired (inherited, purchased, assigned by authorities, loaned, etc), to support the conditional exogeneity of land tenure status, and for the first year in which the parcel was farmed, to account for duration of tenure effects. In columns (1), (3), and (4) pooled regressions are run, clustering standard errors at the parish level. In column (2), farmer-group fixed effects are added to the clustered model. This model is the most conservative, but because of relatively low input use and the geographic clustering of farmer groups (restricting the identifying variation somewhat) results are at times insignificant (although patterns are generally consistent with the pooled models).

6.1 Tenure Security

Before turning to agricultural inputs used on parcels, recall that exogenous land pressures should influence input decisions through two pathways: increasing optimal input usage by increasing the relative price of cash crops (which should be equally true for all parcels), and increased tenure insecurity (attenuating the cropping response, but only for long-term inputs under customary tenure). Customary tenure is not necessarily less secure than freehold; rather, it can become less secure if local elites face incentives to expropriate customary land. These incentives, I argue, increase with local land values.

Actual elite expropriation of customary land to sell to outsiders is relatively rare and therefore difficult to detect statistically in random-sample surveys such as this one. However, even if respondents have not personally experienced elite expropriation, they may worry about it, particularly as their community experiences the external pressures that have led to elite expropriation elsewhere. If this pattern holds, I would expect respondents to express greater concern about losing customary parcels as land pressures increase. These land pressures should not affect the perceived security of freehold land to the same extent, as titles provide some legal recourse in case of expropriation.

Table 1 presents regression results using the average perceived insecurity of a parcel as the outcome. That is, for each parcel, respondents were asked first who in the household had rights to that parcel, and what the likelihood was that each rights-holder would involuntarily lose the land (on a scale of 1 to 5, with higher numbers representing a greater chance of losing it). These responses were averaged across all rights-holders to each parcel to measure household-level insecurity about that parcel. Respondents reported for 21% of parcels that it was ‘slightly likely’ or greater that they would involuntarily lose rights to the parcel, so this is a meaningful concern in our sample.

In table 1, tenure insecurity responds weakly or even negatively to higher land pressures on freehold land, but demonstrates a statistically significant positive relationship with land pressures (measured by rental prices) on customary land. This implies that customary land becomes more insecure than freehold as land pressures rise. These results suggest that the hypothesized mechanism for agricultural investment (rising insecurity on customary land) does indeed hold in Uganda. In table A4, I conduct a similar exercise using a larger sample of data collected for monitoring & evaluation purposes of ACDP across an additional 30 districts (this data does not contain consistent measures of input use, so cannot be used for the full analysis). Results are consistent, even stronger, when using this more geographically diverse sample.

6.2 Crop Choice

In the model, the share of land devoted to cash crops, $(1 - \alpha)$, is taken as exogenous for simplicity. In reality, however, this parameter is a choice made by farmers. If output prices for cash crops increase along with land pressures (given better access to urban markets), we would expect a shift into cash crops in higher land pressure environments. Therefore, in table 2, I regress an indicator for at least one plot on the parcel being primarily planted to a cash crop on my three measures of land pressures. Table A39 instead looks at an analogous measure for staple crops, with roughly the inverse pattern.

Overall, farmers are more likely to plant cash crops in higher land pressure environments, and this relationship holds equally for both freehold and customary parcels. This is unsurprising: nearby urban markets increase demand locally for fragile cash crops such as fruits and vegetables. The model demonstrates how this higher demand for input-intensive cash crops should increase optimal input use for both long-term inputs such as trees as well as short-term ones such as pesticides and inorganic fertilizer. The divergence in long-term input application on customary versus freehold land with increasing land pressures therefore can be attributed directly to insecurity. In the tables presented in the main text, I control for whether or not any plot on the parcel is primarily devoted to a cash crop, to isolate the insecurity effect. However, there may be some concern that this specification suffers from ‘bad control’ problems, as crop choice is potentially an outcome of land pressures; therefore, in Appendix C.7.2, I present regressions that omit this control and show qualitatively similar patterns as the main text. Unfortunately, the ACDP survey does not indicate the specific crop on a plot to which an input was applied if multiple crops are grown (and intercropping is very common in this context), so I cannot separately estimate input use responses on cash crops and staples.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0213 (0.0217)	0.0247 (0.0156)		
Customary $\times$ Rental Price	0.106*** (0.0137)	0.120*** (0.0186)		
Inverse Travel Time			-1.397*** (0.466)	
Customary $\times$ Inverse Travel Time			0.767 (0.501)	
Decrease in Travel Time				-2.651* (1.326)
Customary $\times$ Decrease in Travel Time				-0.186 (1.140)
Customary	0.103 (0.0781)	0.0759 (0.0747)	-0.499** (0.222)	-0.149 (0.199)
Wealth Index	-0.00221 (0.00248)	-0.00164 (0.00268)	-0.00143 (0.00259)	-0.000411 (0.00242)
Landholdings	-0.00440 (0.00278)	-0.00549 (0.00363)	-0.00544** (0.00241)	-0.00592** (0.00268)
Cash Crops	0.0564 (0.0711)	0.00479 (0.0766)	0.0764 (0.0774)	0.104 (0.0877)
First Year Farming	0.00149** (0.000641)	0.00127 (0.000861)	0.00139** (0.000689)	0.00130* (0.000672)
Constant	-1.321 (1.257)	-1.228 (1.738)	-0.846 (1.353)	-0.811 (1.298)
Observations	3,240	3,240	3,222	3,222
R-squared	0.261	0.382	0.259	0.253
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 1: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	Planted Cash Crop			
	(1)	(2)	(3)	(4)
Rental Price	0.0125*	0.0103**		
	(0.00646)	(0.00446)		
Customary $\times$ Rental Price	0.0179	-0.000699		
	(0.0160)	(0.00283)		
Inverse Travel Time			0.750*	
			(0.394)	
Customary $\times$ Inverse Travel Time			0.327	
			(0.297)	
Decrease in Travel Time				5.452***
				(0.741)
Customary $\times$ Decrease in Travel Time				2.551***
				(0.844)
Customary	-0.129***	-0.0152	-0.000478	0.218***
	(0.0313)	(0.0140)	(0.0531)	(0.0623)
Wealth Index	0.00584***	-0.000504	0.00452**	9.75e-05
	(0.00208)	(0.000634)	(0.00173)	(0.00113)
Landholdings	0.00201*	0.000790	0.00224*	0.00268**
	(0.00104)	(0.000880)	(0.00112)	(0.00105)
Observations	5,511	5,511	5,489	5,489
R-squared	0.094	0.389	0.106	0.189
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

*** p<0.01, ** p<0.05, * p<0.1

Table 2: OLS results of a dummy equal to one for parcels with at least one plot primarily devoted to a cash crop on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

6.3 Tree Planting

Planting trees in Uganda represents a long-term investment in agricultural productivity. In an inter-cropped system, trees provide structure for climbing crops (increasing the productivity of those crops), prevent soil erosion, and produce fruit or coffee themselves (Deininger and Ali, 2008), though their value takes some time to be realized. Therefore, the expected benefits of tree planting are lessened as the risk of expropriation increases, as modeled above.

Trees have a more nuanced relationship with tenure security on customary land. Because trees are such a visible investment in the land, planting trees on a parcel can actually reduce the risk of expropriation, particularly by neighbors who understand the social context of customary tenure. Therefore, it might be expected that in more isolated (low land-pressure areas) where local social dynamics dominate, tree-planting might actually be more valuable on customary land than on freehold. As external land pressures increase and the primary security concern shifts to elite expropriation to sell to outsiders, this security-enhancing effect may not be as strong. Therefore, the model would predict that as external land pressures increase, tree-planting on customary land may increase; on freehold land, tree-planting would increase substantially. In the ACDP data, I observe whether or not there is a tree on the parcel; given the long duration of trees, this stock measure may not perfectly capture recent tree-planting investment. That is, trees planted in the more distant past on customary parcels may not have been planted in a new, higher land pressure environment. However, if this does bias my results, it should make it harder to detect a difference in responses between freehold and customary land empirically than is predicted by the model.

When looking at table 3, this pattern does largely hold. When land pressures are measured using standardized rental prices or relative decrease in travel time, there is a small, statistically insignificant increase in tree-planting on freehold land; when using the inverse travel time to Kampala to proxy these pressures, freehold land sees more trees as external pressures mount. With all measures, however, the response is (often statistically significantly) attenuated on customary parcels, as predicted by the model and seen in negative coefficients on the interaction terms.

As a point of interest, note that in figures 4 - 6, tree planting is actually more common on customary plots than freehold at low land values. This stems from the role of trees in establishing tenure in customary systems: trees demonstrate to neighbors effective control and serve as a visible investment in land stewardship, thereby forestalling ‘horizontal’ encroachment. This is less effective when dealing with outsiders to the social system, so the effect reverses as land pressures increase; in this case, trees serve primarily as a long-term investment in the agricultural productivity of the parcel. PPI, our proxy for wealth, actually has a negative relationship with tree-planting; poorer households are more likely to plant trees, perhaps because they serve as a way to

signal ownership if other avenues are unavailable to poor households. In contrast, other inputs (particularly those requiring up-front cash purchases such as commercial fertilizers) are more common among wealthier households, in tables 4 - 6.

6.4 Organic Fertilizer Use

Organic fertilizer use both improves agricultural productivity in the season in which it is applied, and can also have medium-term effects by improving soil organic content (Johansen et al., 2015).⁴⁸ Indeed, Deininger et al. (2021) follow a long literature in treating organic fertilizer application as a key long-term investment in land productivity in the Sub-Saharan African smallholder context, which can be affected by tenure security (Jacoby et al., 2002; Jacoby and Mansuri, 2008). The model would predict, then, that organic fertilizer use would increase along with land values on freehold land, but the effect will be attenuated on customary land.

When looking at a binary indicator for the application of organic fertilizer in Table 4, the relationships are somewhat mixed. Organic fertilizer use does seem to respond positively to land pressures on both freehold and customary land (especially as measured by travel time; the relationship is less statistically significant for standardized rental prices). On customary land, the response is actually stronger than for freehold, although this is not always true for rental prices. Broadly, these results are consistent with the model's predictions for an intermediate-term input that is not immediately visible to potential tenure challengers, but may have substantial returns on cash crops.

In appendix B, I also use the value of organic fertilizer⁴⁹ applied as an outcome, following Fenske (2011)'s suggestion that this is more likely to detect statistically significant relationships with tenure. In these Tobit regressions, results are more sensitive to specification, and the inclusion of farmer group fixed effects makes a large difference.

6.5 Inorganic Fertilizer Use

Inorganic fertilizer, by contrast, largely exhibits short-term returns to agricultural land. The model predicts that for short-term inputs, rising land values will increase the relative price of cash crops and thus increase optimal short-term input use equivalently on freehold and customary land. In table 5, inorganic fertilizer use responds more to land pressures on customary than freehold land, although the differences are small. This is despite the fact that I am controlling for whether or not there is a cash crop on the parcel, and so can be interpreted as isolating the tenure security channel.

⁴⁸Improved soil content can also have complementarities with inorganic fertilizer use, although this is not studied in detail in the current analysis.

⁴⁹Standardized from a shilling value; this data is often imputed as many farmers generate and apply their own organic fertilizer including manure outside of the market.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0127 (0.00817)	0.00844 (0.00954)		
Customary $\times$ Rental Price	-0.0435*** (0.0116)	-0.0377*** (0.00841)		
Inverse Travel Time			0.886** (0.396)	
Customary $\times$ Inverse Travel Time			-7.85e-05 (0.432)	
Decrease in Travel Time				1.007 (0.885)
Customary $\times$ Decrease in Travel Time				-1.703** (0.809)
Customary	0.0283 (0.0433)	0.0233 (0.0507)	0.276** (0.109)	0.299*** (0.0905)
Wealth Index	-0.00162 (0.00153)	0.00167** (0.000821)	-0.00264* (0.00132)	-0.00139 (0.00121)
Landholdings	0.000321 (0.00152)	0.00147 (0.00136)	0.000742 (0.00146)	6.51e-05 (0.00158)
Cash Crop	0.146*** (0.0344)	0.130*** (0.0385)	0.133*** (0.0315)	0.137*** (0.0357)
First Year Farming	-0.00285*** (0.000798)	-0.00324*** (0.00105)	-0.00284*** (0.000808)	-0.00287*** (0.000780)
Constant	6.208*** (1.583)	6.491*** (2.105)	6.011*** (1.584)	6.143*** (1.541)
Observations	3,250	3,250	3,232	3,232
R-squared	0.061	0.242	0.071	0.066
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 3: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for a dummy equal to one if the primary crop on any plot on the parcel is a cash crop, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	-0.000986 (0.00129)	-0.00120 (0.000964)		
Customary $\times$ Rental Price	0.0137*** (0.00508)	0.00307 (0.00199)		
Inverse Travel Time			0.541*** (0.199)	
Customary $\times$ Inverse Travel Time			0.297*** (0.105)	
Decrease in Travel Time				3.411*** (0.445)
Customary $\times$ Decrease in Travel Time				1.192*** (0.310)
Customary	-0.0661** (0.0249)	-0.0184 (0.0210)	0.000714 (0.0419)	0.170*** (0.0452)
Wealth Index	0.00398*** (0.000663)	0.000915 (0.000584)	0.00318*** (0.000666)	0.00100 (0.000627)
Landholdings	0.00102 (0.00114)	0.00127 (0.000985)	0.00177 (0.00126)	0.00242** (0.000973)
Cash Crop	0.207*** (0.0308)	0.0368 (0.0254)	0.194*** (0.0300)	0.137*** (0.0271)
First Year Planting	-1.78e-05 (0.000530)	-0.000113 (0.000471)	-0.000117 (0.000526)	-4.07e-05 (0.000475)
Constant	0.0102 (1.081)	0.197 (0.972)	0.0988 (1.055)	-0.159 (0.958)
Observations	3,248	3,248	3,230	3,230
R-squared	0.169	0.322	0.175	0.224
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 4: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for a dummy equal to one if the primary crop on any plot on the parcel is a cash crop, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

The patterns are also mixed when we look in table A2 at the value of inorganic fertilizer applied for those who do use inorganic fertilizer; application on customary parcels seems more responsive to land pressures particularly as proxied by inverse travel time. This is not inconsistent with the theoretical model's predictions: if long-term input use decreases due to new insecurity, farmers may compensate with additional short-term input use, particularly given the lumpy nature of long-term investments such as trees.

6.6 Pesticide Use

Pesticides, another short-term input, are generally used as-needed in the Ugandan agricultural sector rather than being applied proactively. The model would predict that use of pesticides responds similarly to land pressures on freehold and customary parcels; however, in table 6 there is a mixed pattern on differential responsiveness by tenure type, depending on the measure of land pressures used. In column (2), looking within farmer groups, pesticide use actually diminishes with rental prices on customary parcels; in columns (3) and (4) it increases with inverse travel time or the decrease in travel time on both freehold and customary parcels, and perhaps even more so on customary parcels. These differences are generally smaller than those observed for long-term inputs, as the increasing insecurity on customary land does not impact decisions about pesticide use.

6.7 Thresholds

In the sections above, I have modeled the relationship between land pressures and agricultural investment as linear on both customary and freehold land. It could be, however, that there are multiple regimes of land pressures where the relationship differs. To explore this empirically, I use a threshold of 2 hours travel time to Kampala; in interviews, this was identified as the distance within which most urban residents are seeking agricultural land, and therefore there should be higher likelihood of elite expropriation of customary land in these areas. In appendix C.3, I compare investment outcomes on customary vs. freehold parcels in low vs. high pressure environments using dummy variables and their interactions, for a parsimonious representation of my results. Consistent with the model, I find lower input use on customary parcels overall; there is higher input use closer to Kampala, but input use differs significantly across tenure types close to Kampala.

Beyond this question of whether the levels of investment differ more or less than 2 hours travel time from Kampala, it may be that the responsiveness of investment to changes in land pressures differ. That is, do the slopes in figure 5 differ on either side of a two-hour travel time cutoff? It could be that within this 2-hour commute of Kampala, the insecurity stemming from outside pressures has already been 'priced in' to farmers' investment decisions, and therefore investment behavior on customary vs. freehold parcels is

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00330 (0.00267)	0.00497** (0.00186)		
Customary $\times$ Rental Price	0.00894 (0.00865)	0.00422 (0.00263)		
Inverse Travel Time			0.750*** (0.0846)	
Customary $\times$ Inverse Travel Time			0.588*** (0.108)	
Decrease in Travel Time				1.396*** (0.426)
Customary $\times$ Decrease in Travel Time				0.658** (0.323)
Customary	0.00844 (0.0363)	0.00489 (0.0405)	0.0528 (0.0476)	0.0867** (0.0336)
Wealth Index	0.00456*** (0.000756)	0.00203** (0.000878)	0.00323*** (0.000733)	0.00332*** (0.000856)
Landholdings	0.00168** (0.000789)	0.00148* (0.000863)	0.00279*** (0.000904)	0.00269*** (0.000939)
Cash Crop	0.0151 (0.0147)	0.0143 (0.0175)	0.00318 (0.0132)	-0.0126 (0.0160)
First Year Farming	0.00129*** (0.000471)	0.000900* (0.000532)	0.00122** (0.000462)	0.00133*** (0.000459)
Constant	-2.677*** (0.949)	-1.920* (1.072)	-2.703*** (0.933)	-2.866*** (0.924)
Observations	3,245	3,245	3,227	3,227
R-squared	0.029	0.150	0.057	0.041
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 5: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for a dummy equal to one if the primary crop on any plot on the parcel is a cash crop, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00142 (0.00342)	-0.00205 (0.00253)		
Customary $\times$ Rental Price	0.0116 (0.00812)	-0.0124*** (0.00333)		
Inverse Travel Time			0.855*** (0.118)	
Customary $\times$ Inverse Travel Time			0.584** (0.227)	
Decrease in Travel Time				2.155*** (0.502)
Customary $\times$ Decrease in Travel Time				0.00778 (0.544)
Customary	-0.0315 (0.0657)	-0.0441 (0.0672)	0.0455 (0.0497)	0.192*** (0.0425)
Wealth Index	0.00590*** (0.000705)	0.00335*** (0.000995)	0.00424*** (0.000838)	0.00424*** (0.000961)
Landholdings	0.00151 (0.00144)	0.000662 (0.00100)	0.00205 (0.00129)	0.00184* (0.00107)
Cash Crop	0.145*** (0.0261)	0.0876*** (0.0208)	0.130*** (0.0265)	0.106*** (0.0260)
First Year Farming	0.000623 (0.000483)	0.000547 (0.000552)	0.000533 (0.000469)	0.000606 (0.000487)
Constant	-1.241 (0.953)	-1.175 (1.099)	-1.218 (0.910)	-1.351 (0.933)
Observations	3,244	3,244	3,226	3,226
R-squared	0.079	0.203	0.103	0.096
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 6: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the a dummy equal to one if the primary crop on any plot on the parcel is a cash crop, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

relatively unchanging as land pressures increase within this high-pressure environment. Alternately, in more remote areas, outsider demand for land may not be salient, so customary land is not perceived as any less secure even as land pressures ramp up. To explore this, I separately run the regressions on either side of the cutoff (regression tables in appendix C.4).

Figure 7 shows, like figures 4 - 6, the difference between freehold and customary parcels as land pressures increase, but models this relationship separately on either side of the two-hour travel time threshold (recall that the inverse travel time measure is $(1 + time_j)^{-1}$, so two hours is located at $1/3$ on the x-axis). Interestingly, relative tenure insecurity, which in the previous figures was downwards-sloping (customary parcels became relatively more insecure as land pressures increased) is here upwards-sloping although negative close to Kampala (implying that although customary land is less secure than freehold, the difference narrows closer to Kampala), but downwards-sloping further away. This is in contrast to the various investments, which retain their direction on both sides of the threshold. Perhaps perceptions of tenure insecurity have adapted to external threats close to Kampala. In contrast, investments on customary vs. freehold land diverge more sharply as land pressures increase close to Kampala: note the steeper slopes on the right-hand side of Figure 7, within 2 hours of Kampala. In more remote areas, outsider demand shaping insecurity seems to translate less intensely into patterns of agricultural investment.

This result, in conjunction with the prior linear specifications, has interesting policy implications for where titling may have the greatest benefits. Recall that in general, customary tenure is less secure close to Kampala, where outside demand for land leaves customary parcels vulnerable to elite expropriation. This then leads to an underinvestment in long-term agricultural inputs on customary parcels in these areas. From the threshold analysis, I conclude that investment on customary parcels is more sensitive to land pressures closer to Kampala. Therefore, securing rights to those parcels, such as by titling them and bringing them from the customary realm into freehold, should have larger benefits (such as unlocking agricultural productivity) close to Kampala.

This implies that the government should target titling activities in high land pressure environments, such as those easily accessed from Kampala. Not only are the benefits of clarifying customary rights and making them legally enforceable higher in areas with more demand from outsiders, the costs of titling, including specialized staff such as surveyors, are likely lower in more easily accessible areas. This pattern, where the net benefits of clarifying the potential ambiguity of customary land rights through formal documentation are higher in environments with higher land pressures, likely holds across a variety of customary tenure contexts on the continent. Indeed, there is evidence that formalization has been more successful in unlocking agricultural investment (and faced higher demand) in near-urban contexts in Burkina Faso (Huntington et al., 2023). At the same time, the higher stakes of documenting rights in locations with ready demand for clearly

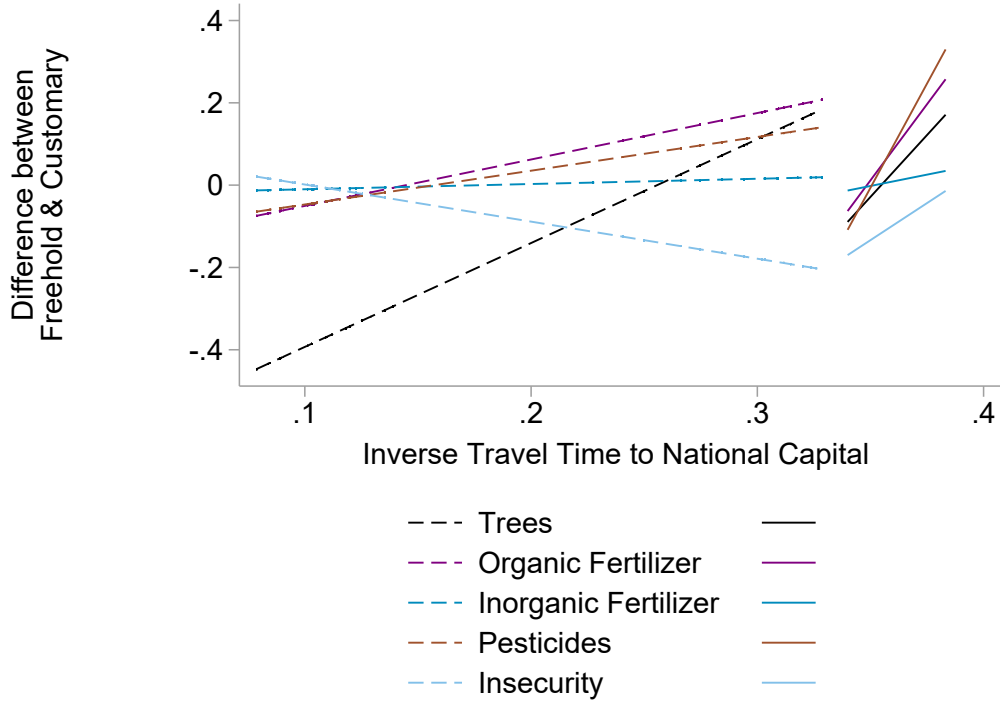


Figure 7: The divergence in input use across customary and freehold parcels is more responsive to land pressures closer to Kampala.

documented land may lead to elite capture of the titling process (Lawry et al., 2023).⁵⁰

7 Conclusion

In this paper, I developed a theoretical model of short- and long-term agricultural investments on customary and freehold land. This model incorporates how expropriation incentives change as land pressures increase. Local elites, observing outsiders' new demand for land and inability to distinguish between overlapping customary rights, may choose to expropriate land used by non-elites in the community to sell. Even if uncommon in practice, the perception and fear of elite expropriation reduces tenure security on customary tenure parcels. This, in turn, reduces the expected returns to long-term inputs.

The model predicts that long-term input use will diverge between customary and freehold parcels as land pressures increase in an area. Indeed, when I test the model's implications using survey data from four regions of Uganda, I find that long-term inputs (such as trees and organic fertilizer) are used more often on freehold land as land pressures increase, while on customary land this relationship is attenuated by tenure

⁵⁰For example, in Burkina Faso, local elections faced more political competition in anticipation of land documentation in municipalities closer to urban centers, consistent with political attempts to control the documentation process (Meinzen-Dick, 2026).

insecurity. Changing cropping patterns due to the proximity of urban markets appears to also influence short-term input use, but the effects on long-term inputs are larger, suggesting an independent effect of tenure security on long-term inputs.

The theoretical model and empirical results together suggest the importance of expanding beyond a simplification of customary tenure as less secure to thinking carefully about the social nature of land rights and the interacting incentives of multiple stakeholders. This insight allows researchers to make more detailed predictions about agricultural investment and may help explain the mixed results in much of the empirical literature. Secure customary tenure may become less secure as tenure systems intersect. This also suggests targeted policy to document and clarify rights in areas vulnerable to elite expropriation as land pressures mount to protect the rights of the most vulnerable. In particular, this work suggests that the net benefits of expensive documentation and formalization of rights will be highest in areas such as those accessible to cities that face high land pressures from outsiders, as the risk of expropriation of customary land is a more binding constraint on investment in those regions.

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A General Analytical Solution

Consider a two-period model, where the farmer has allocated a (fixed) share $0 \leq \alpha \leq 1$ of their exogenously-given land endowment L under tenure system h to staple crops s and the remainder $(1 - \alpha)$ to cash crops c . In the first period, the farmer choose to apply fertilizer, F_1^s and F_1^c , and trees, T^s and T^c , to the staple and cash crops respectively; each crop has its own production function and may have different marginal productivities of inputs. I leave these production functions general, only assuming diminishing marginal returns to each input ($\frac{\partial^2 f}{\partial F^2} < 0$, $\frac{\partial^2 f}{\partial T^2} < 0$). These assumptions allow me to argue that in cases where the marginal productivity of a given input are lower, the level of that input used is higher, and therefore to predict how investment will respond to changing land values, which is the empirically relevant hypothesis, even without estimating a full agronomic production function which is beyond the scope of this model. The production functions for each crop may differ, and in particular may have different marginal productivities of inputs. The farmer also has some wealth endowment, w_1 , and cannot borrow. When considering second period returns, the farmer has a discount rate of β and faces some probability that their land will be expropriated before the second period, in which case fixed investments (trees) would be lost as well. This probability is given by $(1 - \phi(T, h, vL))$ and could be a function of tree investment, tenure system, as well as the value of the land. After working through the main features of the model, I will consider what different forms of this ϕ function could imply for the farmer's decisions.

The first period problem faced by the farmer is then to:

$$\max_{F_1^s, F_1^c, T^s, T^c} \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T^s + T^c, h, vL) \pi_2^*(F_2^s, F_2^c, T^s, T^c) \quad (2)$$

$$\text{subject to: } p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \quad (3)$$

While in the second period, the farmer only chooses fertilizer (F_2^s and F_2^c):

$$\max_{F_2^s, F_2^c} \alpha p_s f_s(F_2^s, T^s) + (1 - \alpha) p_c f_c(F_2^c, T^c) - p_F(F_2^s + F_2^c) \quad (4)$$

$$\begin{aligned} \text{subject to: } & p_F(F_2^s + F_2^c) \leq w_2 \\ & w_2 = w_1 \end{aligned} \quad (5)$$

The First Order Conditions of this second stage are:

$$\alpha p_s \frac{\partial f_s}{\partial F_2^s} - p_F - \lambda_1 p_F = 0 \quad (6)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_2^c} - p_F - \lambda_1 p_F = 0 \quad (7)$$

$$\lambda_1 [p_F(F_2^s + F_2^c) - w_1 - \alpha p_s f_s(F_1^s, T^s) - (1 - \alpha) p_c f_c(F_1^c, T^c) + p_F(F_1^s + F_1^c) + p_T(T^s + T^c)] = 0 \quad (8)$$

Consider the case where the budget constraint does not bind in the second stage, as the constraint binding does not add substantively to understanding the mechanism at play. Therefore, $\lambda_1 = 0$.

The equilibrium conditions are then:

F_2^{s*} such that:

$$\frac{\partial f_s}{\partial F_2^s} = \frac{p_F}{\alpha p_s} \quad (9)$$

and F_2^{c*} such that:

$$\frac{\partial f_c}{\partial F_2^c} = \frac{p_F}{(1 - \alpha) p_c} \quad (10)$$

Therefore

$$\pi_2^* = \alpha p_s f_s(F_2^{s*}, T^s) + (1 - \alpha) p_c f_c(F_2^{c*}, T^c) - p_F(F_2^{s*} + F_2^{c*}) \quad (11)$$

and the derivatives with respect to the first period choice variables are:

$$\frac{\partial \pi_2^*}{\partial F_1^s} = 0 \quad (12)$$

$$\frac{\partial \pi_2^*}{\partial F_1^c} = 0 \quad (13)$$

$$\frac{\partial \pi_2^*}{\partial T^s} = \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} \quad (14)$$

$$\frac{\partial \pi_2^*}{\partial T^c} = (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} \quad (15)$$

Turning to the first stage, then:

$$\max_{F_1^s, F_1^c, T^s, T^c} \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T_1^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T^s, T^c, h, vL) \pi_2^*(F_2^s, F_2^c, T^s, T^c) \quad (16)$$

$$\text{subject to: } p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \quad (17)$$

The First Order Conditions are then:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial F_1^s} - \lambda_2 p_F = 0 \quad (18)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial F_1^c} - \lambda_2 p_F = 0 \quad (19)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial T^s} - \lambda_2 p_T = 0 \quad (20)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial T^c} - \lambda_2 p_T = 0 \quad (21)$$

$$\lambda_2 [p_F(F_1^s + F_1^c) + p_T(T^s + T^c) - w_1] = 0 \quad (22)$$

Which simplify to:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F - \lambda_2 p_F = 0 \quad (23)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F - \lambda_2 p_F = 0 \quad (24)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} - \lambda_2 p_T = 0 \quad (25)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} - \lambda_2 p_T = 0 \quad (26)$$

$$\lambda_2 [p_F(F_1^s + F_1^c) + p_T(T^s + T^c) - w_1] = 0 \quad (27)$$

Consider a farmer who has a sufficient initial endowment of wealth w_1 that the budget constraint does not bind in either period ($\lambda_1 = \lambda_2 = 0$)

The first-period FOCs are then:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F = 0 \quad (28)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F = 0 \quad (29)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} = 0 \quad (30)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} = 0 \quad (31)$$

This then gives us:

$$\frac{\partial f_s}{\partial F_1^s} = \frac{p_F}{\alpha p_s} \quad (32)$$

$$\frac{\partial f_c}{\partial F_1^c} = \frac{p_F}{(1 - \alpha) p_c} \quad (33)$$

Which are the same conditions as in the second period. Therefore, in the totally unconstrained case, $F_1^{s*} = F_2^{s*}$ and $F_1^{c*} = F_2^{c*}$. Furthermore, if I assume that under freehold tenure, $\phi = 1$ and $\frac{\partial \phi}{\partial T} = 0$ (tenure is perfectly secure and planting trees has no effect on tenure security), then

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} = 0 \quad (34)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} = 0 \quad (35)$$

Because $F_1^{s*} = F_2^{s*}$ and $F_1^{c*} = F_2^{c*}$ in this totally unconstrained case, $\frac{\partial f}{\partial T} = \frac{\partial f}{\partial T} \Big|_{F_2^*}$. The equilibrium conditions are then

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T}{\alpha p_s (1 + \beta)} \quad (36)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T}{(1 - \alpha) p_c (1 + \beta)} \quad (37)$$

However, if $\phi < 1$, such as is possible under customary tenure, and $\frac{\partial \phi}{\partial T} > 0$ then the equilibrium conditions are:

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{\alpha p_s (1 + \beta \phi)} \quad (38)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{(1 - \alpha) p_c (1 + \beta \phi)} \quad (39)$$

Compared to the freehold case, the inclusion of $\phi < 1$ increases the marginal productivity of trees, lowering optimal investment in this long term input; this effect is attenuated by the security-enhancing effect of tree planting ($\frac{\partial \phi}{\partial T} > 0$, therefore $p_T > p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*$). The model therefore accords with most theoretical models of tenure and long-term investments.

Let us now consider how optimal allocation would respond to a change in land pressures. As land pressures increase, particularly if this is driven by greater access to urban markets, we would expect the relative price of the cash crop p_c to increase. That is, compared to the prices of inputs and the staple crop output price, the price farmers can receive for their cash crops such as fruits and vegetables is higher with new access to relatively wealthy urban markets.

As p_c increases, $\frac{\partial f_c}{\partial F_1^c}$ decreases, which implies the optimal fertilizer use on cash crops increases. Furthermore, on freehold parcels, $\frac{\partial f_c}{\partial T^c}$ also would decrease as seen in equation (37), so optimal tree planting on cash crops increases. Optimal input use on staple crops will not changed for farmers whose wealth constraint does not bind.

On customary parcels, in equation (39), p_c will increase; importantly, however, the likelihood of expropriation from local elites who hold transfer rights to customary parcels also increases. In the model, this can be captured as $\frac{\partial \phi}{\partial v} < 0$: ϕ , the expected likelihood of retaining a parcel and its returns in the second period, decreases with rising land pressures. This ‘vertical’ expropriation does not recognize investments made in the land such as trees; assume for simplicity that ‘horizontal’ expropriation, such as border incursions, is constant in its responsiveness to tree investment, so $\frac{\partial \phi}{\partial T}$ is constant as land pressures rise. Therefore, referring to the equilibria conditions for customary parcels (equations (38) and (39), repeated here):

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{\alpha p_s (1 + \beta \phi)} \quad (40)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{(1 - \alpha) p_c (1 + \beta \phi)} \quad (41)$$

We can see that on staple crops, the reduction in ϕ increases $\frac{\partial f_s}{\partial T^s}$, implying that optimal tree planting on staple crops will decrease. On cash crops, this impact of a reduction in ϕ increasing $\frac{\partial f_c}{\partial T^c}$ works against

the increase in p_c , which decreases the equilibrium marginal productivity of trees; the net effect of rising land pressures on tree planting for cash crops on customary parcels will depend on the relative magnitude of these two mechanisms.

This final point is the key effect of the model. Contrary to responses on freehold land, which simply increase optimal long- and short-term investment in cash crops, on customary land the impacts are also affected by changing tenure security. On staple crops, optimal long-term investment actually decreases. On cash crops, the response is theoretically ambiguous, but certainly lower than on freehold parcels.

B Continuous Measures of Input Use

Fenske (2011), in his quantitative review of existing evidence on tenure and agricultural investment in West Africa, finds that using binary measures of investment limits identifying variation and makes it more likely to find insignificant results. Above, I do find statistically significant relationships between tenure and binary investment outcomes, particularly when I explicitly consider the land pressure environment. Nevertheless, the ACDP survey's detailed agricultural production module allows me to construct continuous measures of long- and short-term investment: the value of fertilizer or pesticides applied. Unfortunately, I cannot construct a similar measure for the most long-term of my inputs, tree planting.

Due to the relatively rare use of agricultural inputs, I adapt Fenske (2011)'s suggestion to use the trimmed LAD estimator to my cross-sectional data and use Tobit estimation. These results can be interpreted as in a linear model, although the linear effect represented by the coefficient is the slope on the uncensored (non-zero) outcome. Tables A1 - A3 are generally consistent with the binary outcome results presented in the main text, with a greater divergence between customary and freehold parcels on longer-term inputs (organic fertilizer) than short-term inputs. However, I do not prioritize these results due to the lack of precision in reported valuations of inputs, which in the ACDP survey do not accord with those from other sources in Uganda.

VARIABLES	Value Organic Fertilizer Used			
	(1)	(2)	(3)	(4)
Rental Price	0.0948 (0.0925)	-0.0593*** (0.0146)		
Customary $\times$ Rental Price	0.416 (0.308)	-21.12*** (0.622)		
Inverse Travel Time			19.02** (9.309)	
Customary $\times$ Inverse Travel Time			4.714 (5.807)	
Decrease in Travel Time				85.78** (39.12)
Customary $\times$ Decrease in Travel Time				52.33** (24.63)
Customary	-1.034** (0.463)	0.0520 (0.142)	3.197 (2.919)	3.474 (3.234)
Wealth Index	0.200* (0.110)	0.117*** (0.00346)	0.179* (0.101)	0.141* (0.0814)
Landholdings	0.0231 (0.0268)	0.101*** (0.0131)	0.0365 (0.0241)	0.0533** (0.0219)
Cash Crop	3.720** (1.834)	0.394*** (0.117)	3.334* (1.772)	1.803* (1.069)
First Year Farming	0.0303 (0.0260)	0.0310*** (9.76e-05)	0.0268 (0.0246)	0.0256 (0.0233)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

*** p<0.01, ** p<0.05, * p<0.1

Table A1: Tobit results with the outcome of the value of organic fertilizer applied to a parcel (a medium-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
	Value Inorganic Fertilizer Used			
Rental Price	0.0263*** (0.00540)	0.0255*** (0.000310)		
Customary $\times$ Rental Price	0.0188 (0.0341)	-0.532*** (0.0191)		
Inverse Travel Time			2.972*** (1.032)	
Customary $\times$ Inverse Travel Time			2.130* (1.125)	
Decrease in Travel Time				3.527 (2.520)
Customary $\times$ Decrease in Travel Time				1.599 (1.849)
Customary	0.103 (0.132)	0.0719*** (0.0127)	0.345 (0.342)	0.310 (0.237)
Wealth Index	0.0177*** (0.00626)	0.0105*** (0.000320)	0.0131** (0.00573)	0.0145** (0.00587)
Landholdings	-0.00290 (0.00450)	-0.00831*** (0.00156)	0.000238 (0.00547)	-0.000250 (0.00493)
Cash Crop	-0.0176 (0.0598)	0.0238 (0.0186)	-0.0154 (0.0574)	-0.0673 (0.0824)
First Year Farming	0.00612* (0.00321)	0.00531*** (7.70e-06)	0.00631* (0.00324)	0.00668** (0.00327)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A2: Tobit results with the outcome of the value of inorganic fertilizer applied to a parcel (a short-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Value Pesticides Used		
Rental Price	0.0935 (0.128)	0.0520*** (0.00260)		
Customary $\times$ Rental Price	0.136 (0.137)	-0.497*** (0.0205)		
Inverse Travel Time			9.216 (5.914)	
Customary $\times$ Inverse Travel Time			19.33 (13.99)	
Decrease in Travel Time				22.47* (12.66)
Customary $\times$ Decrease in Travel Time				4.450 (8.374)
Customary	-0.327 (0.754)	-0.344*** (0.117)	-3.381 (2.837)	1.700 (1.297)
Wealth Index	0.0681*** (0.0249)	0.0517*** (0.00118)	0.0492*** (0.0158)	0.0497*** (0.0154)
Landholdings	-0.00741 (0.0179)	-0.0325*** (0.00466)	-0.00643 (0.0205)	-0.0116 (0.0237)
Cash Crop	1.773** (0.718)	0.874*** (0.0358)	1.707** (0.744)	1.393** (0.587)
First Year Farming	0.00469 (0.00559)	0.00880*** (3.18e-05)	0.00184 (0.00646)	0.00350 (0.00566)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A3: Tobit results with the outcome of the value of pesticides applied to a parcel (a medium-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C Robustness Checks

C.1 Expanded Geographic Sample

The baseline survey for the impact evaluation of ACDP was conducted in four districts in Uganda, with an extensive survey including all agricultural inputs and outputs. However, a more concise survey was additionally conducted for monitoring and evaluation (M&E) purposes in an additional 30 districts located around the country. This M&E survey only asks about agricultural investments on a single target crop in each district (rice, beans, maize, cassava, or coffee), and is therefore not well-suited for the main analysis conducted above. Nevertheless, it does contain parcel-level tenure characteristics, such as perceived insecurity and tenure system, as well as rental prices and GPS locations that allow me to match with travel time data as above. Therefore, I am able to repeat the exercise from table 1 using this larger, more geographically diverse sample in table A4. The results are, if anything, even stronger: tenure insecurity is unrelated to rising land pressures on freehold land, but customary parcels are felt more insecure in higher land pressure environments.

VARIABLES	Average Insecurity			
	(1)	(2)	(3)	(4)
Rental Price	-0.00121 (0.00110)	0.000555 (0.00128)		
Customary $\times$ Rental Price	5.639*** (0.687)	5.079*** (1.097)		
Inverse Travel Time			-0.306 (0.271)	
Customary $\times$ Inverse Travel Time			1.113*** (0.237)	
Decrease in Travel Time				0.453 (0.308)
Customary $\times$ Decrease in Travel Time				0.580** (0.258)
Customary	0.0766** (0.0345)	0.0660* (0.0358)	-0.302*** (0.0734)	0.00312 (0.0388)
Wealth Index	2.44e-05 (0.000973)	-0.000557 (0.000984)	-0.000797 (0.00105)	-0.000688 (0.000987)
Landholdings	2.83e-07 (2.49e-07)	2.54e-06*** (3.07e-07)	4.92e-07** (2.31e-07)	4.32e-07* (2.49e-07)
Observations	10,253	10,229	10,283	10,283
R-squared	0.256	0.331	0.256	0.251
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

*** p<0.01, ** p<0.05, * p<0.1

Table A4: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Data comes not only from the Impact Evaluation sample used throughout the paper, but also from the more condensed ACDP Monitoring and Evaluation survey conducted in an additional 30 districts.

C.2 Spatial Correlation

In my primary specification above, I cluster standard errors at the parish level to account for any geographic relationships between households in the same parish. Here, I model the spatial dependencies more explicitly, using Conley (1999)’s spatially dependent standard errors. I allow for spatial dependence up to 50km, which would include all households within a district. These regressions, as in the main paper, include controls for how the parcel was acquired, plus the shown wealth index and total household landholdings. These results are the most different of all my robustness checks. In particular, in table A6, the coefficient on Customary $\times$ Inverse Travel Time becomes negative and statistically significant (at the 10% level), when in most other specifications (such as table 3) it is positive but insignificant. However, this is if anything more consistent with the theoretical model, where long-term input use is attenuated on customary parcels as land pressures rise. Similarly, the coefficient on Customary $\times$ Decrease in Travel Time switching from positive and significant in table 4 to negative and significant in table A7 is more in line with the model’s predictions of attenuation on customary parcels for this medium-term input. On the other hand, the same pattern is observed for pesticides in table A9, which is slightly more difficult to explain but may be due to spatial correlation in pest damage that is accounted for in this estimation.

VARIABLES	(1)	(2)	(3)	(4)
	Average Insecurity			
Rental Price	0.0213 (0.0227)	0.0247 (0.0170)		
Customary $\times$ Rental Price	0.0851*** (0.0203)	0.0953*** (0.00829)		
Inverse Travel Time			-1.397*** (0.368)	
Customary $\times$ Inverse Travel Time			2.164*** (0.630)	
Decrease in Travel Time				-2.651* (1.485)
Customary $\times$ Decrease in Travel Time				2.465 (2.230)
Customary	0.103 (0.0988)	0.0759 (0.0692)	-0.499*** (0.149)	-0.149 (0.159)
Wealth Index	-0.00221 (0.00141)	-0.00164 (0.00222)	-0.00143 (0.000916)	-0.000411 (0.000419)
Landholdings	-0.00440 (0.00380)	-0.00549* (0.00330)	-0.00544* (0.00326)	-0.00592* (0.00326)
Cash Crop	0.0564 (0.0680)	0.00479 (0.0727)	0.0764 (0.0845)	0.104 (0.0916)
First Year Farming	0.00149*** (0.000536)	0.00127** (0.000609)	0.00139*** (0.000489)	0.00130*** (0.000450)
Constant	-1.321 (1.146)	-1.228 (1.393)	-0.846 (1.065)	-0.811 (1.004)
Observations	3,240	3,240	3,222	3,222
R-squared	0.261	0.382	0.259	0.253
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A5: GMM results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0127 (0.00950)	0.00844 (0.00907)		
Customary $\times$ Rental Price	-0.0563*** (0.00731)	-0.0461*** (0.00923)		
Inverse Travel Time			0.886 (0.563)	
Customary $\times$ Inverse Travel Time			-0.886*** (0.289)	
Decrease in Travel Time				1.007 (1.286)
Customary $\times$ Decrease in Travel Time				-2.710** (1.183)
Customary	0.0283 (0.0758)	0.0233 (0.0739)	0.276*** (0.0319)	0.299*** (0.0985)
Wealth Index	-0.00162 (0.00171)	0.00167** (0.000722)	-0.00264 (0.00183)	-0.00139 (0.00142)
Landholdings	0.000321 (0.00169)	0.00147 (0.00126)	0.000742 (0.00175)	6.51e-05 (0.00171)
Cash Crop	0.146** (0.0595)	0.130*** (0.0404)	0.133*** (0.0372)	0.137*** (0.0477)
First Year Farming	-0.00285*** (0.00103)	-0.00324** (0.00163)	-0.00284** (0.00117)	-0.00287*** (0.00107)
Constant	6.208*** (1.980)	6.491** (3.270)	6.011*** (2.195)	6.143*** (1.986)
Observations	3,250	3,250	3,232	3,232
R-squared	0.061	0.242	0.071	0.066
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A6: GMM results of a dummy equal to one for parcels with trees planted on them (a long term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	-0.000986 (0.00105)	-0.00120 (0.000761)		
Customary $\times$ Rental Price	0.0147** (0.00607)	0.00428** (0.00170)		
Inverse Travel Time			0.541 (0.371)	
Customary $\times$ Inverse Travel Time			-0.245 (0.314)	
Decrease in Travel Time				3.411*** (0.885)
Customary $\times$ Decrease in Travel Time				-2.219*** (0.535)
Customary	-0.0661 (0.0544)	-0.0184 (0.0211)	0.000714 (0.0327)	0.170*** (0.0511)
Wealth Index	0.00398*** (0.00110)	0.000915*** (0.000341)	0.00318*** (0.000861)	0.00100** (0.000428)
Landholdings	0.00102 (0.00140)	0.00127 (0.00120)	0.00177 (0.00164)	0.00242* (0.00131)
Cash Crop	0.207*** (0.0189)	0.0368*** (0.00782)	0.194*** (0.0189)	0.137*** (0.0170)
First Year Farming	-1.78e-05 (0.000233)	-0.000113 (0.000662)	-0.000117 (0.000246)	-4.07e-05 (0.000265)
Constant	0.0102 (0.509)	0.197 (1.382)	0.0988 (0.458)	-0.159 (0.505)
Observations	3,248	3,248	3,230	3,230
R-squared	0.169	0.322	0.175	0.224
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A7: GMM results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Inorganic Fertilizer	(3)	(4)
Rental Price	0.00330*** (0.000359)	0.00497*** (0.000932)		
Customary $\times$ Rental Price	0.00564 (0.00632)	-0.000753 (0.00164)		
Inverse Travel Time			0.750*** (0.0896)	
Customary $\times$ Inverse Travel Time			-0.162 (0.0994)	
Decrease in Travel Time				1.396*** (0.513)
Customary $\times$ Decrease in Travel Time				-0.739* (0.418)
Customary	0.00844 (0.0445)	0.00489 (0.0478)	0.0528** (0.0227)	0.0867*** (0.0328)
Wealth Index	0.00456*** (0.000850)	0.00203*** (0.000546)	0.00323*** (0.00106)	0.00332*** (0.00125)
Landholdings	0.00168*** (0.000557)	0.00148*** (0.000304)	0.00279*** (0.000742)	0.00269*** (0.000553)
Cash Crop	0.0151 (0.0144)	0.0143*** (0.00524)	0.00318 (0.0125)	-0.0126** (0.00598)
First Year Farming	0.00129** (0.000541)	0.000900 (0.000606)	0.00122** (0.000551)	0.00133*** (0.000475)
Constant	-2.677** (1.119)	-1.920 (1.238)	-2.703** (1.132)	-2.866*** (0.994)
Observations	3,245	3,245	3,227	3,227
R-squared	0.029	0.150	0.057	0.041
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A8: GMM results of a dummy equal to one for parcels with inorganic fertilizer applied (a short term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00142 (0.000955)	-0.00205*** (0.000473)		
Customary $\times$ Rental Price	0.0130* (0.00697)	-0.0103* (0.00530)		
Inverse Travel Time			0.855*** (0.0652)	
Customary $\times$ Inverse Travel Time			-0.271 (0.334)	
Decrease in Travel Time				2.155*** (0.453)
Customary $\times$ Decrease in Travel Time				-2.147*** (0.758)
Customary	-0.0315 (0.0973)	-0.0441 (0.0990)	0.0455** (0.0229)	0.192*** (0.0547)
Wealth Index	0.00590*** (0.000723)	0.00335** (0.00141)	0.00424*** (0.00106)	0.00424*** (0.00139)
Landholdings	0.00151 (0.00150)	0.000662* (0.000385)	0.00205* (0.00118)	0.00184** (0.000823)
Cash Crop	0.145*** (0.0201)	0.0876*** (0.00684)	0.130*** (0.0106)	0.106*** (0.0146)
First Year Farming	0.000623** (0.000257)	0.000547 (0.000445)	0.000533* (0.000276)	0.000606** (0.000304)
Constant	-1.241** (0.490)	-1.175 (0.874)	-1.218** (0.545)	-1.351** (0.580)
Observations	3,244	3,244	3,226	3,226
R-squared	0.079	0.203	0.103	0.096
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A9: GMM results of a dummy equal to one for parcels with pesticides applied (a short term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C.3 Splitting Sample

The relationship between land pressures and input use on customary vs. freehold land may not be linear, as modeled in my primary specifications. Therefore, I also look at binary splits in the various land pressure measures, and compare outcomes on customary vs. freehold parcels in low vs. high land pressure environments using interactions.

First, in table A10, I create a dummy variable equal to one for all parcels that have an estimated rental price per acre above the median rental price in their district, and consider those parcels to be facing higher land pressures. The results accord with those above: input use is similar on customary and freehold parcels in low-land pressure areas; higher land pressure parcels are more likely to have inputs applied; but this effect is attenuated on customary parcels. I check, in table A11, that these results are not just driven by the distribution of customary vs freehold parcels across low vs. high land value areas, but the distribution is relatively similar.

Then, in table A12, I compare across districts, considering those close to Kampala (Masaka & Iganga) as facing high land pressures and those further away (Amuru & Ntungamo) having lower land pressures. This distinction is somewhat similar to the prior threshold of two hours travel time that I use in my threshold analysis. Here again, results are consistent with my model: input use is generally less common on customary parcels, more likely in districts near Kampala, but the difference between customary and freehold parcels is significantly different in districts near Kampala. Once again, customary and freehold parcels are distributed across both sets of districts, as seen in table A13.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Trees		Organic Fertilizer		Inorganic Fertilizer		Pesticides		Average Insecurity	
Customary	0.0888** (0.0382)	0.0695* (0.0394)	-0.0392* (0.0228)	0.00539 (0.0163)	0.0157 (0.0395)	0.0148 (0.0425)	0.0259 (0.0455)	0.0211 (0.0405)	0.154* (0.0791)	0.113 (0.0868)
Higher Price	0.0805** (0.0385)	0.0922** (0.0446)	0.0812*** (0.0222)	0.0193 (0.0161)	-0.0535*** (0.0165)	-0.0330* (0.0179)	0.0192 (0.0180)	0.0235* (0.0134)	0.0803 (0.104)	0.0644 (0.0943)
Customary $\times$ Higher Price	-0.131** (0.0539)	-0.105* (0.0606)	-0.0556** (0.0222)	-0.0542* (0.0277)	-0.0263 (0.0330)	-0.0282 (0.0296)	-0.123*** (0.0401)	-0.137*** (0.0431)	-0.0814 (0.113)	-0.0574 (0.1000)
Constant	6.355*** (1.655)	6.783*** (2.210)	0.154 (1.034)	0.317 (0.973)	-2.735*** (0.984)	-2.020* (1.131)	-1.146 (0.964)	-0.896 (1.162)	-0.989 (1.221)	-0.871 (1.707)
Observations	3,268	3,268	3,266	3,266	3,263	3,263	3,262	3,262	3,258	3,258
R-squared	0.063	0.244	0.174	0.323	0.037	0.152	0.084	0.206	0.252	0.372
Fixed Effect	None	Group	None	Group	None	Group	None	Group	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A10: OLS results where the outcome in columns (1) - (8) is a dummy equal to one if the given input was applied to the parcel, and in columns (9) - (10) is the average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure). Customary is a dummy equal to one for customary parcels and zero otherwise. Higher price is a dummy equal to one if the parcel has an estimated rental price per acre above the median rental price in the district. Customary $\times$ Higher price is the interaction of these two dummies. Controls (not shown) include the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2), (4), (6), (8), and (10), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Differences in number of observations between outcomes is due to missing data for some outcomes.

Tenure Status	Low Rental Price	High Rental Price	Total
Freehold	2,003	1,905	3,908
Customary	1,203	900	2,103
Total	3,206	2,805	6,011

Table A11: Customary and Freehold parcels are found in relatively similar proportions in low and high rental value environments.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Trees		Organic Fertilizer	Inorganic Fertilizer			Pesticides		Average Insecurity	
Customary	0.190** (0.0825)	0.0884* (0.0509)	-0.0319 (0.0272)	-0.00359 (0.0261)	0.0419 (0.0317)	0.0180 (0.0276)	0.0209 (0.0369)	-0.0169 (0.0314)	-0.349** (0.160)	-0.453** (0.170)
Near Kampala	0.258*** (0.0571)	0.577*** (0.0203)	0.101*** (0.0296)	-0.00561 (0.0110)	0.152*** (0.0172)	-0.0294 (0.0208)	0.160*** (0.0314)	-0.0157 (0.0207)	-0.365*** (0.0864)	1.064*** (0.0389)
Customary $\times$ Near Kampala	-0.184* (0.0911)	-0.0774 (0.0650)	-0.0397 (0.0345)	-0.0185 (0.0218)	-0.0402* (0.0206)	-0.0191 (0.0284)	-0.0579 (0.0528)	-0.0299 (0.0511)	0.521*** (0.152)	0.611*** (0.170)
Constant	6.016*** (1.674)	6.415*** (2.068)	-0.0198 (1.028)	0.187 (0.942)	-2.786*** (0.966)	-2.068* (1.128)	-1.344 (0.906)	-1.174 (1.067)	-0.820 (1.343)	0.230 (1.836)
Observations	3,268	3,268	3,266	3,266	3,263	3,263	3,262	3,262	3,258	3,258
R-squared	0.087	0.240	0.174	0.322	0.055	0.149	0.099	0.201	0.266	0.380
Fixed Effect	None	Group	None	Group	None	Group	None	Group	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A12: OLS results where the outcome in columns (1) - (8) is a dummy equal to one if the given input was applied to the parcel, and in columns (9) - (10) is the average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure). Customary is a dummy equal to one for customary parcels and zero otherwise. Near Kampala is a dummy equal to one if the parcel is in Masaka or Iganga districts, which are closer to Kampala. Customary $\times$ Near Kampala is the interaction of these two dummies. Controls (not shown) include the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2), (4), (6), (8), and (10), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Differences in number of observations between outcomes is due to missing data for some outcomes.

Tenure Status	Far from Kampala	Near Kampala	Total
Freehold	835	3,948	4,783
Customary	962	1,364	2,326
Total	1,797	5,312	7,109

Table A13: Customary and Freehold parcels are distributed across districts near and far from Kampala, although freehold is relatively more common near Kampala than further away.

C.4 Threshold Analysis

In this section, I present the regression results looking separately on either side of a threshold of 2 hours travel time from Kamapala (identified in interviews as being a reasonable radius for external demand for land).

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Average Insecurity			
Rental Price	0.0174 (0.0202)	-0.108 (0.464)				
Customary $\times$ Rental Price	0.101*** (0.0155)	10.34*** (1.463)				
Inverse Travel Time			-1.697*** (0.546)	-2.278 (5.110)		
Customary $\times$ Inverse Travel Time			0.709 (0.714)	-4.989 (6.236)		
Decrease in Travel Time					-2.093 (1.504)	-3.951* (1.902)
Customary $\times$ Decrease in Travel Time					-0.549 (1.394)	-4.084** (1.505)
Observations	2,387	817	2,405	817	2,405	817
R-squared	0.323	0.141	0.319	0.131	0.310	0.138
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala
Robust standard errors in parentheses						
*** p<0.01, ** p<0.05, * p<0.1						

Table A14: Regression of average tenure insecurity across all rightholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Trees			
Rental Price	0.0123 (0.00797)	0.0785 (0.176)				
Customary $\times$ Rental Price	-0.0395*** (0.0113)	1.461 (0.902)				
Inverse Travel Time			1.382*** (0.438)	2.264 (3.986)		
Customary $\times$ Inverse Travel Time			0.302 (0.597)	-14.69*** (4.586)		
Decrease in Travel Time					1.224 (0.912)	1.000 (0.946)
Customary $\times$ Decrease in Travel Time					-1.508 (1.061)	-3.195* (1.735)
Observations	2,399	815	2,417	815	2,417	815
R-squared	0.061	0.099	0.086	0.116	0.066	0.105
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A15: Regression of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Organic Fertilizer			
Rental Price	-0.000891 (0.00116)	0.0408 (0.127)				
Customary $\times$ Rental Price	0.0157*** (0.00412)	-0.265 (0.554)				
Inverse Travel Time			0.952*** (0.278)	11.30*** (1.503)		
Customary $\times$ Inverse Travel Time			0.403*** (0.126)	6.951*** (1.593)		
Decrease in Travel Time					3.960*** (0.562)	4.967*** (0.770)
Customary $\times$ Decrease in Travel Time					1.278*** (0.341)	2.156* (1.168)
Observations	2,396	816	2,414	816	2,414	816
R-squared	0.184	0.149	0.206	0.220	0.248	0.236
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A16: Regression of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Inorganic Fertilizer			
Rental Price	0.00353 (0.00289)	0.0632 (0.0990)				
Customary $\times$ Rental Price	0.00882 (0.00650)	-1.498** (0.536)				
Inverse Travel Time			0.694*** (0.135)	0.318 (1.792)		
Customary $\times$ Inverse Travel Time			0.559*** (0.152)	-4.182 (2.912)		
Decrease in Travel Time					1.388** (0.588)	-0.238 (0.608)
Customary $\times$ Decrease in Travel Time					0.561 (0.392)	-1.711 (1.237)
Constant	-2.529** (1.000)	-1.144 (2.063)	-2.796** (1.038)	-1.164 (2.191)	-2.815*** (0.966)	-1.248 (2.020)
Observations	2,396	813	2,414	813	2,414	813
R-squared	0.031	0.027	0.055	0.027	0.043	0.027
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A17: Regression of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Pesticides			
Rental Price	-0.00102 (0.00354)	0.116 (0.134)				
Customary $\times$ Rental Price	0.00811 (0.00574)	-0.211 (1.072)				
Inverse Travel Time			0.877*** (0.179)	4.075 (2.695)		
Customary $\times$ Inverse Travel Time			0.369 (0.293)	-10.94** (3.907)		
Decrease in Travel Time					2.062*** (0.618)	1.308 (1.020)
Customary $\times$ Decrease in Travel Time					-0.331 (0.558)	-5.531*** (1.383)
Constant	-0.732 (1.081)	-2.194 (2.782)	-1.095 (1.034)	-3.554 (2.890)	-1.006 (0.961)	-2.566 (2.652)
Observations	2,394	814	2,412	814	2,412	814
R-squared	0.103	0.060	0.123	0.076	0.120	0.084
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A18: Regression of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

C.5 Parish Fixed Effects

My preferred specification makes use of the farmer group recruitment structure of the ACDP data to account for unobserved heterogeneity in farmer quality, which is likely to be shared among comembers of a self-selected farmer group. However, here I also present regressions using the more traditional administrative unit (parish in this case, given inconsistent identification of villages in the data) fixed effects. Results are consistent with those presented in the body of the paper.

VARIABLES	(1) Trees	(2) Used Organic	(3) Used Inorganic	(4) Used Pesticides	(5) Insecurity
Rental Price	0.0120 (0.00982)	3.30e-05 (0.000662)	0.00283 (0.00255)	-0.00274 (0.00290)	0.0218 (0.0203)
Customary $\times$ Rental Price	-0.0420*** (0.00716)	0.00338 (0.00214)	-0.00765*** (0.00260)	-0.00773** (0.00312)	0.113*** (0.0176)
Customary	0.0325 (0.0504)	-0.0220 (0.0241)	0.00680 (0.0403)	-0.0431 (0.0687)	0.104 (0.0814)
Wealth Index	0.000590 (0.000949)	0.00101* (0.000541)	0.00291*** (0.000951)	0.00393*** (0.000778)	-0.00124 (0.00248)
Landholdings	0.000376 (0.00168)	0.00137 (0.000988)	0.00189** (0.000725)	0.00169 (0.00128)	-0.00550 (0.00335)
Cash Crop	0.136*** (0.0396)	0.0390 (0.0276)	0.0210 (0.0216)	0.0980*** (0.0202)	0.0464 (0.0821)
First Year Farming	-0.00353*** (0.00100)	-8.47e-05 (0.000391)	0.00109** (0.000454)	0.000559 (0.000467)	0.00167** (0.000729)
Constant	7.936*** (2.025)	0.223 (0.816)	-2.266** (0.927)	-1.153 (0.935)	-1.527 (1.438)
Observations	3,250	3,248	3,245	3,244	3,240
R-squared	0.156	0.298	0.088	0.152	0.297
Fixed Effect	Parish	Parish	Parish	Parish	Parish
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A19: OLS results of a dummy for the given input applied to the parcel (in columns (1) - (4) or average tenure security across all rightsholders on the parcel (scale of 1-5, with higher value indicating more insecure, reported in column (5)) on standardized rental prices for freehold and customary parcels, controlling for the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Parish fixed effects are included in all specifications, and standard errors are clustered at the parish level for all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise.

C.6 Alternate Measures

C.6.1 Validating Perceptions of Insecurity

Arnot et al. (2011) discusses how different proxies for tenure security have been used (to varying effect) in the literature. The ACDP survey directly elicits how worried respondents are (on a scale of 1-5) about losing their land in the next 5 years, which I average over all rightsholders on a given parcel as my primary measure of tenure insecurity. However, I also show in table A20 that these perceptions are related to common sources of insecurity. Having a formal document issued by a land board (either a title, lease, or certificate of customary ownership, in most cases) is associated with lower levels of insecurity, as household rights are more easily legally enforceable with such a document. On the other hand, parcels which have in the past been cause for concern over conflict or are currently involved in disputes are (unsurprisingly) held to be more insecure. Although these correlates of perceived insecurity are not the primary focus of this paper, they are consistent with findings in the literature and validate these survey measures.

VARIABLES	Average Insecurity					
	(1)	(2)	(3)	(4)	(5)	(6)
Formal Document	-0.289*** (0.0646)	-0.0901 (0.0631)				
Ever concerned dispute			0.936*** (0.140)	0.872*** (0.0860)		
Active conflicts					1.277*** (0.179)	1.120*** (0.138)
Wealth Index		-0.00178 (0.00215)		-0.00324 (0.00207)		-0.00193 (0.00236)
Landholdings		-0.00506* (0.00278)		-0.00658* (0.00335)		-0.00561* (0.00294)
First Year Farming		0.00119* (0.000673)		0.000985 (0.000795)		0.00104 (0.000738)
Constant	1.494*** (0.0483)	-0.632 (1.294)	1.255*** (0.0276)	-0.314 (1.554)	1.320*** (0.0292)	-0.458 (1.412)
Observations	5,589	3,293	5,591	3,296	5,596	3,302
R-squared	0.031	0.249	0.129	0.337	0.075	0.296
Controls	None	Parcel Access	None	Parcel Access	None	Parcel Access
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A20: OLS results of an average of tenure security across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel of measures of land pressures. Columns (2), (4) and (6) control for the Progress out of Poverty wealth index, total household landholdings, the first year the parcel was farmed, and the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications. Difference in number of observations between columns are due to missing data for some variables.

C.6.2 Alternate Measures of Land Values

Given that parcel rental prices may be endogenous to soil quality or past agricultural investment, I also explore alternate measures of land values. First, I predict per-acre rental prices using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. This should strip away noise in the estimated rental prices, and follows the intuition of a hedonic model for land values. Second, I use the median per-acre rental price in a village, to eliminate parcel-level heterogeneity in rental values. Finally, I use a non-standardized version of per-acre rental prices, which exhibits considerable skewness, but can be interpreted as the shilling value. Results with these three alternatives in tables C.6.2 - C.6.2 are by and large consistent with those in the main text, although are generally noisier and therefore less likely to be statistically significant.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Average Insecurity		
Predicted Price	2.17e-07 (1.91e-07)	4.85e-07** (2.37e-07)			
Customary $\times$ Predicted Price	3.32e-07 (4.69e-07)	7.72e-07 (5.44e-07)			
Village Price			-3.64e-06*** (8.59e-07)		
Customary $\times$ Village Price			1.22e-06 (9.75e-07)		
Rental Price (UGX)				4.10e-08 (4.16e-08)	4.74e-08 (3.00e-08)
Customary $\times$ Rental Price (UGX)				2.04e-07*** (2.62e-08)	2.30e-07*** (3.58e-08)
Customary	0.0731 (0.174)	-0.0129 (0.146)	-0.802*** (0.295)	0.0713 (0.0790)	0.0399 (0.0766)
Wealth Index	-0.00276 (0.00260)	-0.00145 (0.00284)	-0.00142 (0.00244)	-0.00221 (0.00248)	-0.00164 (0.00268)
Landholdings	-0.00503* (0.00273)	-0.00634* (0.00374)	-0.00630** (0.00249)	-0.00440 (0.00278)	-0.00549 (0.00363)
Cash Crop	0.0415 (0.0686)	0.0252 (0.0715)	0.0887 (0.0732)	0.0564 (0.0711)	0.00479 (0.0766)
First Year Farming	0.00135** (0.000641)	0.000752 (0.000836)	0.00138* (0.000696)	0.00149** (0.000641)	0.00127 (0.000861)
Observations	3,256	3,256	3,258	3,240	3,240
R-squared	0.253	0.376	0.267	0.261	0.382
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A21: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3) Trees	(4)	(5)
Predicted Price	3.97e-07** (1.74e-07)	5.56e-07** (2.69e-07)			
Customary Predicted Price	-3.33e-07*** (1.03e-07)	-7.77e-08 (1.77e-07)			
Village Price			2.33e-06*** (5.81e-07)		
Customary $\times$ Village Price			4.25e-07 (8.27e-07)		
Rental Price (UGX)				2.44e-08 (1.57e-08)	1.62e-08 (1.83e-08)
Customary $\times$ Rental Price (UGX)				-8.36e-08*** (2.23e-08)	-7.24e-08*** (1.61e-08)
Customary	0.216*** (0.0557)	0.179*** (0.0521)	0.385*** (0.140)	0.0496 (0.0444)	0.0407 (0.0520)
Wealth Index	-0.00159 (0.00147)	0.00171** (0.000828)	-0.00285** (0.00130)	-0.00162 (0.00153)	0.00167** (0.000821)
Landholdings	0.000507 (0.00151)	0.00170 (0.00139)	0.00147 (0.00143)	0.000321 (0.00152)	0.00147 (0.00136)
Cash Crop	0.129*** (0.0334)	0.122*** (0.0373)	0.124*** (0.0290)	0.146*** (0.0344)	0.130*** (0.0385)
First Year Farming	-0.00290*** (0.000806)	-0.00346*** (0.00107)	-0.00279*** (0.000812)	-0.00285*** (0.000798)	-0.00324*** (0.00105)
Observations	3,266	3,266	3,268	3,250	3,250
R-squared	0.075	0.255	0.083	0.061	0.242
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A22: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Organic Fertilizer		
Predicted Price	1.43e-07 (1.21e-07)	-6.49e-08 (6.39e-08)			
Customary $\times$ Predicted Price	1.24e-07** (5.66e-08)	-7.14e-08 (6.28e-08)			
Village Price			9.57e-07*** (2.94e-07)		
Customary $\times$ Village Price			5.14e-07*** (1.69e-07)		
Rental Price (UGX)				-1.89e-09 (2.47e-09)	-2.31e-09 (1.85e-09)
Customary $\times$ Rental Price (UGX)				2.63e-08*** (9.76e-09)	5.90e-09 (3.82e-09)
Customary	-0.0677 (0.0473)	-0.0158 (0.0386)	0.0162 (0.0541)	-0.0716*** (0.0260)	-0.0200 (0.0210)
Wealth Index	0.00397*** (0.000703)	0.000885 (0.000563)	0.00344*** (0.000658)	0.00398*** (0.000663)	0.000915 (0.000584)
Landholdings	0.00166 (0.00135)	0.00156 (0.00105)	0.00199 (0.00135)	0.00102 (0.00114)	0.00127 (0.000985)
Cash Crop	0.193*** (0.0296)	0.0276 (0.0253)	0.194*** (0.0312)	0.207*** (0.0308)	0.0368 (0.0254)
First Year Farming	-6.17e-05 (0.000529)	-8.66e-05 (0.000458)	-4.75e-06 (0.000510)	-1.78e-05 (0.000530)	-0.000113 (0.000471)
Observations	3,264	3,264	3,266	3,248	3,248
R-squared	0.168	0.323	0.174	0.169	0.322
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A23: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Inorganic Fertilizer		
Predicted Price	-3.49e-08 (6.92e-08)	-7.30e-08 (8.26e-08)			
Customary × Predicted Price	-9.15e-08** (3.90e-08)	-1.62e-07** (6.99e-08)			
Village Price			1.47e-06*** (1.74e-07)		
Customary × Village Price			9.68e-07*** (1.78e-07)		
Rental Price (UGX)				6.34e-09 (5.12e-09)	9.54e-09** (3.57e-09)
Customary × Rental Price (UGX)				1.72e-08 (1.66e-08)	8.10e-09 (5.05e-09)
Customary	0.0226 (0.0327)	0.0289 (0.0355)	0.100*** (0.0344)	0.00631 (0.0377)	0.00518 (0.0409)
Wealth Index	0.00470*** (0.000740)	0.00192** (0.000897)	0.00347*** (0.000753)	0.00456*** (0.000756)	0.00203** (0.000878)
Landholdings	0.00205* (0.00102)	0.00191* (0.00106)	0.00304*** (0.000899)	0.00168** (0.000789)	0.00148* (0.000863)
Cash Crop	0.0174 (0.0145)	0.00289 (0.0222)	-0.000165 (0.0143)	0.0151 (0.0147)	0.0143 (0.0175)
First Year Farming	0.00131*** (0.000484)	0.00102* (0.000555)	0.00136*** (0.000470)	0.00129*** (0.000471)	0.000900* (0.000532)
Observations	3,261	3,261	3,263	3,245	3,245
R-squared	0.030	0.151	0.054	0.029	0.150
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A24: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Pesticides		
Predicted Price	1.81e-07** (7.12e-08)	6.75e-08 (8.47e-08)			
Customary $\times$ Predicted Price	-3.80e-08 (1.18e-07)	-1.37e-07 (1.12e-07)			
Village Price			1.57e-06*** (3.02e-07)		
Customary $\times$ Village Price			8.57e-07** (3.75e-07)		
Rental Price (UGX)				-2.73e-09 (6.56e-09)	-3.93e-09 (4.87e-09)
Customary $\times$ Rental Price (UGX)				2.23e-08 (1.56e-08)	-2.37e-08*** (6.40e-09)
Customary	0.0235 (0.0513)	0.0120 (0.0537)	0.104** (0.0483)	-0.0364 (0.0673)	-0.0402 (0.0679)
Wealth Index	0.00579*** (0.000762)	0.00344*** (0.00101)	0.00471*** (0.000842)	0.00590*** (0.000705)	0.00335*** (0.000995)
Landholdings	0.00158 (0.00133)	0.000513 (0.000953)	0.00236* (0.00139)	0.00151 (0.00144)	0.000662 (0.00100)
Cash Crop	0.135*** (0.0269)	0.0919*** (0.0212)	0.129*** (0.0248)	0.145*** (0.0261)	0.0876*** (0.0208)
First Year Farming	0.000597 (0.000493)	0.000524 (0.000561)	0.000680 (0.000460)	0.000623 (0.000483)	0.000547 (0.000552)
Observations	3,260	3,260	3,262	3,244	3,244
R-squared	0.083	0.202	0.100	0.079	0.203
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A25: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

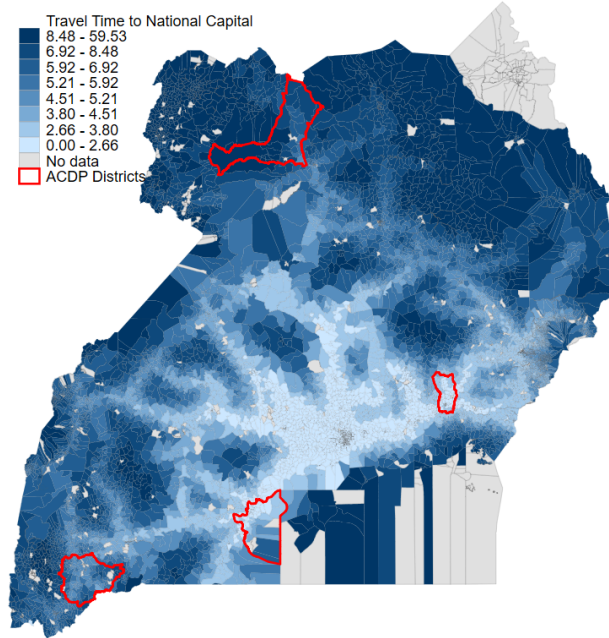


Figure 8: Map of average travel time in 2015 to Kampala for each Parish in Uganda, with ACDP districts highlighted in red. Administrative units in Uganda extend into Lake Victoria to the national border, to include several small islands.

C.6.3 Alternate Measures of Market Access

The external land pressures modeled above (and mapped in figures 8 and 9 are not exclusively coming from the national capital, Kampala. In fact, rapidly-growing regional capitals can create the sort of outsider demand for land that causes such insecurity over customary parcels. Therefore, in tables A26 - A30, I use four more regional measures of land pressures, all from Muller-Crepon (2021). First, I construct analogues to the inverse travel time and 10-year decrease in travel time, but to the regional capital rather than Kampala, the national capital and largest city in the country. However, these regional capitals have changed somewhat dramatically in the last 15 years; Uganda has experienced administrative unit proliferation (Grossman and Lewis, 2014). These newly-named regional capitals, which could drastically improve connections to a ‘regional capital’ in the data, may not reflect the demand for rural land by outsiders that I am modeling. Instead of capturing growing markets and real demand, this may be an artefact of politically-driven fragmentation.

I then also use a measure of market access that Muller-Crepon (2021) provides, stemming from Donaldson (2018). This ‘market access’ measure is a “reduced form expression derived from general equilibrium trade theory” (Donaldson and Hornbeck, 2016), which uses the within-country expansion of transport networks. I use this measure as provided in 2015, as well as the 10-year change in market access. This transport-focused market access, however, is more likely to affect agricultural total factor productivity and is less closely linked to the population-based drivers of external demand for land that I model.

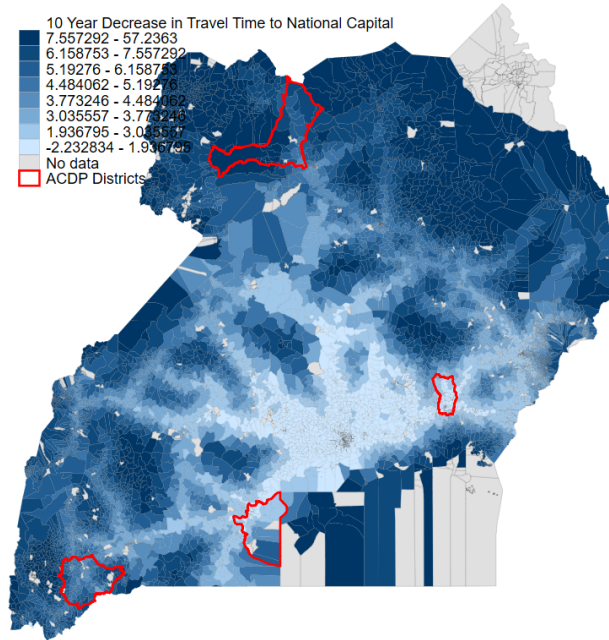


Figure 9: Map of the 10 year change in average travel time in 2015 to Kampala for each Parish in Uganda, with ACDP districts highlighted in red. Administrative units in Uganda extend into Lake Victoria to the national border, to include several small islands.

However, in tables A26 - A30, results with these alternate measures of land pressures are broadly consistent with those in the main text.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Inverse Travel Time to Regional Capital	-0.446*			
	(0.228)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.250			
	(0.234)			
Decrease in Travel Time to Regional Capital		-0.287		
		(0.426)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.133		
		(0.185)		
2015 Market Access			-5.18e-07**	
			(2.18e-07)	
Customary $\times$ 2015 Market Access			4.92e-07	
			(3.49e-07)	
Change in Market Access				0.0333
				(0.111)
Customary $\times$ Change in Market Access				-0.358**
				(0.156)
Customary	-0.239	0.117	-0.0579	0.135**
	(0.200)	(0.0791)	(0.106)	(0.0663)
Wealth Index	-0.00141	-0.00207	-0.00157	-0.00226
	(0.00261)	(0.00256)	(0.00264)	(0.00242)
Landholdings	-0.00529**	-0.00474*	-0.00518*	-0.00561*
	(0.00258)	(0.00275)	(0.00269)	(0.00299)
Cash Crop	0.0647	0.0631	0.0505	0.0742
	(0.0764)	(0.0788)	(0.0747)	(0.0836)
First Year Farming	0.00141**	0.00151**	0.00146**	0.00125*
	(0.000649)	(0.000663)	(0.000623)	(0.000626)
Observations	3,222	3,169	3,222	3,222
R-squared	0.253	0.255	0.254	0.251
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A26: OLS results of average tenure security across all rightholders (scale of 1-5, with higher values indicating more insecure) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Inverse Travel Time to Regional Capital	0.255 (0.200)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.0524 (0.239)			
Decrease in Travel Time To Regional Capital		0.523** (0.225)		
Customary $\times$ Decrease in Travel Time to Regional Capital		0.310* (0.154)		
2015 Market Access			6.95e-08 (2.00e-07)	
Customary $\times$ 2015 Market Access			-3.87e-08 (2.63e-07)	
Change in Market Access				0.0680 (0.0822)
Customary $\times$ Change in Market Access				-0.366*** (0.0651)
Customary	0.128 (0.107)	0.0194 (0.0429)	0.0437 (0.0627)	0.0546** (0.0246)
Wealth Index	-0.00235* (0.00139)	-0.000469 (0.00142)	-0.00170 (0.00136)	-0.00148 (0.00150)
Landholdings	0.000460 (0.00156)	-0.000282 (0.00169)	0.000435 (0.00152)	0.000248 (0.00156)
Cash Crop	0.142*** (0.0326)	0.121*** (0.0326)	0.147*** (0.0368)	0.156*** (0.0394)
First Year Farming	-0.00292*** (0.000903)	-0.00319*** (0.000881)	-0.00278*** (0.000865)	-0.00280*** (0.000795)
Observations	3,232	3,179	3,232	3,232
R-squared	0.063	0.073	0.058	0.067
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A27: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Inverse Travel Time to Regional Capital	0.170*			
	(0.0966)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.126**			
	(0.0550)			
Decrease in Travel Time to Regional Capital		0.309**		
		(0.133)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.00168		
		(0.0499)		
2015 Market Access			2.51e-08	
			(1.05e-07)	
Customary $\times$ 2015 Market Access			6.79e-08	
			(8.66e-08)	
Change in Market Access				0.312***
				(0.0605)
Customary $\times$ Change in Market Access				0.148**
				(0.0587)
Customary	-0.0456	-0.0745***	-0.0754***	-0.0365*
	(0.0386)	(0.0246)	(0.0274)	(0.0190)
Wealth Index	0.00349***	0.00451***	0.00398***	0.00319***
	(0.000700)	(0.000799)	(0.000761)	(0.000508)
Landholdings	0.00146	0.00137	0.00145	0.00137
	(0.00120)	(0.00134)	(0.00131)	(0.00114)
Cash Crop	0.200***	0.189***	0.203***	0.135***
	(0.0297)	(0.0322)	(0.0307)	(0.0346)
First Year Farming	-0.000206	8.14e-06	-7.55e-05	-0.000108
	(0.000533)	(0.000531)	(0.000524)	(0.000447)
Observations	3,230	3,177	3,230	3,230
R-squared	0.169	0.175	0.163	0.201
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A28: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Inverse Travel Time to Regional Capital	0.303*** (0.0622)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.273*** (0.0803)			
Decrease in Travel Time to Regional Capital		0.00591 (0.134)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.120** (0.0513)		
2015 Market Access			2.37e-07** (9.45e-08)	
Customary $\times$ 2015 Market Access			4.33e-07*** (1.20e-07)	
Change in Market Access				0.0153 (0.0704)
Customary $\times$ Change in Market Access				-0.140*** (0.0348)
Customary	0.0231 (0.0618)	0.00571 (0.0386)	-0.0209 (0.0487)	0.0154 (0.0277)
Wealth Index	0.00353*** (0.000740)	0.00445*** (0.000704)	0.00373*** (0.000747)	0.00473*** (0.000788)
Landholdings	0.00230** (0.000964)	0.00239** (0.00107)	0.00230** (0.000994)	0.00215** (0.000995)
Cash Crop	0.0108 (0.0124)	0.0145 (0.0146)	0.0229* (0.0132)	0.0205 (0.0232)
First Year Farming	0.00102** (0.000479)	0.00151*** (0.000466)	0.00111** (0.000476)	0.00130*** (0.000465)
Observations	3,227	3,174	3,227	3,227
R-squared	0.051	0.033	0.044	0.034
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A29: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Inverse Travel Time to Regional Capital	0.353*** (0.0734)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.341** (0.136)			
Decrease in Travel Time to Regional Capital		0.115 (0.161)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.127 (0.0859)		
2015 Market Access			2.55e-07** (1.20e-07)	
Customary $\times$ 2015 Market Access			6.20e-07*** (1.93e-07)	
Change in Market Access				0.123 (0.0986)
Customary $\times$ Change in Market Access				-0.339*** (0.0498)
Customary	-0.0238 (0.0780)	-0.0391 (0.0654)	-0.0859 (0.0783)	0.00247 (0.0319)
Wealth Index	0.00445*** (0.000896)	0.00544*** (0.000802)	0.00467*** (0.000956)	0.00574*** (0.000622)
Landholdings	0.00155 (0.00118)	0.00149 (0.00148)	0.00159 (0.00137)	0.00123 (0.00127)
Cash Crop	0.139*** (0.0270)	0.147*** (0.0237)	0.153*** (0.0266)	0.142*** (0.0230)
First Year Farming	0.000281 (0.000477)	0.000681 (0.000507)	0.000384 (0.000483)	0.000573 (0.000505)
Constant	-0.673 (0.934)	-1.366 (0.993)	-0.774 (0.946)	-1.164 (0.983)
Observations	3,226	3,174	3,226	3,226
R-squared	0.100	0.081	0.094	0.096
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A30: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

C.6.4 Probability of Urbanization

As an alternate proxy for the external pressures on land that I model, I considered using data from the Global Grid of Probabilities of Urban Expansion to 2030 (Seto et al., 2012, 2016). For each 2.5 arc-minute⁵¹ grid cell that was non-urban in 2000, the data include an estimated probability of becoming urban by the year 2030, forecast using a population density driver map. This map is primarily driven by patterns of urbanization, and thus is reasonably exogenous to local agricultural patterns of land values; instead, it isolates the dimension of land values that will also create pressure on local elites to expropriate and sell customary land to outsiders.

However, this data is remarkably bimodal in Uganda, with sparse support of values of the probability of urbanization in the four districts where the baseline study occurred. Thirty-eight percent of parcels fall in a grid cell with a 100% probability of urbanization; another 19% have a 99% probability. There is another grouping at very low levels of predicted urbanization, including the entire northern district of Amuru which has a probability of zero, and then only a few parcels are assigned a probability between .5 and .8. This makes the probability of urbanization not my preferred proxy, but I present regression results with this measure in tables A31 and A32.

I match households to a grid cell using GPS locations collected during the survey (taken at the household residence, not at agricultural fields which could be at some remove). This measure is correlated with the per-acre rental values households reported for their land; the correlation is low across the whole sample (.0450), but when I winsorize several outliers in estimated rental price (among those who own their land and are estimating how much they could rent it out for), this correlation increases to .1492. When I look instead at the median rental price in a given district, the correlation with the probability of urbanization jumps to .6556,⁵² implying (as expected) that this measure is more closely related to broader trends than individual parcel values which may be influenced by soil quality or other agricultural value considerations. Results in table A31 for binary outcomes and A32 for continuous Tobit estimation are relatively consistent with those obtained using either normalized rental values, inverse travel time to Kampala, or the 10-year reduction in travel times, as shown above.

⁵¹2.5 arc-minutes contain approximately 5km at the equator, which Uganda spans

⁵²Similarly, the correlation with median village rental prices is .6697.

VARIABLES	(1) Trees	(2) Used Organic	(3) Used Inorganic	(4) Used Pesticides	(5) Insecurity
Urbanization	0.123** (0.0555)	0.0785** (0.0334)	0.0513 (0.0340)	0.101*** (0.0325)	-0.0787 (0.0765)
Customary $\times$ Urbanization	0.0507 (0.0763)	0.0185 (0.0207)	0.0246 (0.0328)	0.0245 (0.0303)	0.00713 (0.103)
Customary	0.0899 (0.0706)	-0.0165 (0.0250)	0.0297 (0.0407)	0.0340 (0.0529)	0.0432 (0.127)
Wealth Index	-0.00212 (0.00148)	0.00385*** (0.000692)	0.00439*** (0.000813)	0.00550*** (0.000727)	-0.00221 (0.00237)
Landholdings	0.000986 (0.00146)	0.00171 (0.00132)	0.00236** (0.000951)	0.00181 (0.00137)	-0.00562* (0.00291)
Cash Crop	0.137*** (0.0304)	0.198*** (0.0297)	0.0102 (0.0142)	0.137*** (0.0247)	0.0649 (0.0754)
First Year Farming	-0.00292*** (0.000826)	-9.28e-05 (0.000513)	0.00127** (0.000481)	0.000550 (0.000476)	0.00145** (0.000654)
Observations	3,266	3,264	3,261	3,260	3,256
R-squared	0.065	0.169	0.032	0.086	0.251
Fixed Effect	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A31: OLS results where the outcome in columns (1) - (4) is a dummy equal to one if the given input was applied to the parcel, and in column (5) is the average tenure insecurity across all rightsholders. Customary is a dummy equal to one for customary parcels and zero otherwise. The probability of urbanization is a proxy for land pressures, and the regression controls for the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses).

VARIABLES	(1) Value Organic	(2) Value Inorganic	(3) Value Pesticides
Urbanization	3.807 (2.458)	0.128 (0.158)	0.787** (0.383)
Customary $\times$ Urbanization	1.252 (1.537)	0.319* (0.177)	-0.854 (0.987)
Customary	1.369 (1.863)	-0.0662 (0.187)	0.970 (1.132)
Wealth Index	0.190* (0.103)	0.0163*** (0.00616)	0.0680** (0.0267)
Landholdings	0.0390 (0.0282)	0.000153 (0.00403)	-0.00949 (0.0216)
Cash Crop	3.349* (1.710)	-0.0163 (0.0537)	1.740** (0.728)
First Year Farming	0.0282 (0.0251)	0.00654** (0.00325)	0.00499 (0.00542)
Observations	3,270	3,270	3,270
Fixed Effect	None	None	None
Cluster	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A32: Tobit results where the outcome is the value of the given input applied to the parcel. Customary is a dummy equal to one for customary parcels and zero otherwise. The probability of urbanization is a proxy for land pressures, and the regression controls for the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses).

C.7 Additional Controls

C.7.1 Parcel Characteristics

As can be seen in table A33, customary and freehold parcels are not observably identical. In particular, freehold parcels are more likely to have been purchased, more likely to be documented, and less likely to be primarily residential (none of which are surprising). Additionally, we can see different patterns of land use and tenure security: freehold parcels were acquired slightly more recently, are more likely to have formal documents, and are more likely to be irrigated – but are also more likely to have active conflicts and worries about ownership disputes, and less likely to be fallowed in the past two years or practice crop rotation.

Several of these characteristics may be responses to the insecurity incentives modeled in the paper, but I also run regressions controlling for residential land use and a dummy for whether the parcel was purchased (in addition to the controls used throughout the analysis). Although these characteristics are correlated with input use, the primary results of the paper (diverging responses to land pressures for long-term inputs) remain.

C.7.2 Crop Change

One notable divergence between the model and empirical results is the patterns of short-term input use increasing somewhat as land pressures increase, which I argued is likely due to changing in cropping patterns. In particular, many cash crops in Uganda are perennials, meaning that the likelihood of expropriation also affects the returns from these crops; at the same time, many cash crops require higher input use (pesticides in particular) in order to receive a market premium. However, I argue that the divergence in long-term input use between customary and freehold parcels as land pressures increase is not due entirely to changing cropping patterns; there is an additional tenure security effect. This additional marginal impact of tenure security is what is captured in the regressions in the main paper, which control for whether there is a cash crop on the parcel. However, because the shift to cash crops could be a result of the mechanisms of interest, I also run the regressions without this control in tables C.7.2 - A44. Results are, once again, consistent with those in the main text. Table A39 parallels table 2, except with the outcome being a dummy equal to one if there is a staple crop as the primary crop on at least on plot on the parcel (note that in an intercropped system, both cash crops and staple crops can be planted on a parcel). Table A45 focuses specifically on perennial crops as its outcome.

Variable	(1) Freehold		(2) Customary		(1)-(2) Pairwise t-test	
	N	Mean/(SE)	N	Mean/(SE)	N	Difference
Residential Land Use	3854	0.080 (0.004)	2326	0.104 (0.006)	6180	-0.024***
Agricultural Land Use	3854	0.912 (0.005)	2326	0.891 (0.006)	6180	0.021***
Forest Land Use	3854	0.006 (0.001)	2326	0.003 (0.001)	6180	0.003*
Parcel was purchased or rented in	3863	0.739 (0.007)	2326	0.107 (0.006)	6189	0.632***
First Year Farmed, cleaned	2594	2007.114 (0.248)	1023	2001.629 (0.479)	3617	5.486***
Document issued by land board	4023	0.534 (0.008)	2109	0.359 (0.010)	6132	0.175***
Active conflicts	4029	0.040 (0.003)	2106	0.027 (0.004)	6135	0.014***
Ever concerned dispute ownership	4026	0.138 (0.005)	2105	0.066 (0.005)	6131	0.073***
Seasons parcel fallowed in last 2 years	3500	0.195 (0.009)	1947	0.388 (0.017)	5447	-0.193***
Crop rotation practiced on parcel	3540	0.567 (0.008)	2092	0.625 (0.011)	5632	-0.058***
Irrigation on parcel (binary)	3542	0.022 (0.002)	2093	0.013 (0.003)	5635	0.008**

Significance: ***=.01, **=.05, *=.1.

Table A33: Balance table of plot characteristics between freehold and customary parcels. Note that some of these, such as fallowing and crop rotation, are other kinds of investment in the land and therefore may be responding to the incentives described throughout. Residential, agricultural, and forest land use are dummies for the primary use of the parcel. Parcel was purchased or rented in is a dummy equal to one if the respondent reported accessing the parcel through purchase or a formal rental arrangement. Document issued by land board is a dummy equal to one if the parcel has a document (including a title deed, certificate of ownership, certificate of hereditary acquisition, lease or rental contract). Active conflicts is a dummy equal to one if the parcel has an active conflict or dispute ongoing. Ever concerned dispute ownership is a dummy equal to one if the respondent has ever been concerned that someone would dispute their ownership of the parcel. Seasons parcel fallowed in last 2 years is a count of the number of seasons the parcel has been left fallow, out of four potential seasons. Crop rotation and irrigation are dummies equal to one if the respondent reports rotating crops or irrigation on any plot within the parcel.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0213 (0.0216)	0.0247 (0.0156)		
Customary $\times$ Rental Price	0.106*** (0.0136)	0.120*** (0.0187)		
Inverse Travel Time			-1.403*** (0.470)	
Customary $\times$ Inverse Travel Time			0.753 (0.502)	
Decrease in Travel Time				-2.702** (1.331)
Customary $\times$ Decrease in Travel Time				-0.198 (1.139)
Residential	0.0613 (0.0944)	0.00612 (0.0758)	0.0706 (0.0972)	0.0794 (0.0957)
Purchased	0.357 (0.470)	0.375 (0.466)	0.312 (0.486)	0.375 (0.493)
Customary	0.105 (0.0792)	0.0761 (0.0758)	-0.496** (0.222)	-0.151 (0.197)
Wealth Index	-0.00216 (0.00246)	-0.00158 (0.00268)	-0.00138 (0.00260)	-0.000329 (0.00243)
Landholdings	-0.00433 (0.00277)	-0.00551 (0.00362)	-0.00537** (0.00241)	-0.00585** (0.00266)
Cash Crop	0.0533 (0.0714)	0.00459 (0.0767)	0.0733 (0.0775)	0.101 (0.0876)
First Year Farming	0.00155** (0.000627)	0.00126 (0.000837)	0.00146** (0.000678)	0.00137** (0.000665)
o.d6_10	-	-	-	-
Constant	-1.514 (1.201)	-1.299 (1.677)	-1.040 (1.307)	-1.029 (1.254)
Observations	3,240	3,240	3,222	3,222
R-squared	0.262	0.382	0.259	0.253
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A34: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0128 (0.00819)	0.00870 (0.00966)		
Customary $\times$ Rental Price	-0.0438*** (0.0117)	-0.0376*** (0.00839)		
Inverse Travel Time			0.869** (0.400)	
Customary $\times$ Inverse Travel Time			-0.0246 (0.432)	
Decrease in Travel Time				0.928 (0.886)
Customary $\times$ Decrease in Travel Time				-1.700** (0.797)
Residential	0.155** (0.0609)	0.141** (0.0616)	0.148** (0.0617)	0.149** (0.0647)
Purchased	-0.0647 (0.181)	-0.0745 (0.190)	-0.0571 (0.183)	-0.0702 (0.189)
Customary	0.0322 (0.0411)	0.0270 (0.0495)	0.282** (0.107)	0.294*** (0.0890)
Wealth Index	-0.00163 (0.00153)	0.00173** (0.000800)	-0.00259* (0.00131)	-0.00133 (0.00119)
Landholdings	0.000451 (0.00152)	0.00156 (0.00133)	0.000853 (0.00145)	0.000182 (0.00156)
Cash Crop	0.141*** (0.0340)	0.126*** (0.0370)	0.129*** (0.0314)	0.133*** (0.0353)
First Year Farming	-0.00271*** (0.000802)	-0.00309*** (0.00106)	-0.00271*** (0.000819)	-0.00274*** (0.000788)
Observations	3,250	3,250	3,232	3,232
R-squared	0.066	0.245	0.076	0.070
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A35: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	-0.000960 (0.00128)	-0.00105 (0.00102)		
Customary $\times$ Rental Price	0.0135*** (0.00493)	0.00312 (0.00201)		
Inverse Travel Time			0.531** (0.200)	
Customary $\times$ Inverse Travel Time			0.280** (0.105)	
Decrease in Travel Time				3.364*** (0.435)
Customary $\times$ Decrease in Travel Time				1.190*** (0.315)
Residential	0.105*** (0.0384)	0.0798 (0.0503)	0.0975** (0.0387)	0.0839** (0.0379)
Purchased	0.0858 (0.0838)	-0.0505 (0.0697)	0.115 (0.103)	0.0662 (0.113)
Customary	-0.0636*** (0.0226)	-0.0163 (0.0193)	0.00499 (0.0406)	0.167*** (0.0446)
Wealth Index	0.00400*** (0.000697)	0.000951 (0.000601)	0.00322*** (0.000696)	0.00105 (0.000642)
Landholdings	0.00111 (0.00117)	0.00132 (0.000996)	0.00185 (0.00129)	0.00249** (0.00101)
Cash Crop	0.204*** (0.0303)	0.0347 (0.0265)	0.191*** (0.0298)	0.135*** (0.0274)
First Year Farming	7.30e-05 (0.000529)	-2.84e-05 (0.000453)	-2.93e-05 (0.000524)	3.37e-05 (0.000472)
Observations	3,248	3,248	3,230	3,230
R-squared	0.174	0.325	0.180	0.227
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A36: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00328 (0.00267)	0.00484** (0.00191)		
Customary $\times$ Rental Price	0.00903 (0.00875)	0.00417 (0.00264)		
Inverse Travel Time			0.757*** (0.0833)	
Customary $\times$ Inverse Travel Time			0.598*** (0.111)	
Decrease in Travel Time				1.428*** (0.419)
Customary $\times$ Decrease in Travel Time				0.657* (0.328)
Residential	-0.0503 (0.0341)	-0.0695* (0.0348)	-0.0611* (0.0313)	-0.0586* (0.0343)
Purchased	0.0910 (0.183)	0.0281 (0.184)	0.0595 (0.180)	0.0312 (0.181)
Customary	0.00710 (0.0379)	0.00308 (0.0423)	0.0504 (0.0487)	0.0886** (0.0334)
Wealth Index	0.00457*** (0.000763)	0.00199** (0.000865)	0.00322*** (0.000750)	0.00329*** (0.000865)
Landholdings	0.00164** (0.000793)	0.00143 (0.000864)	0.00274*** (0.000903)	0.00265*** (0.000939)
Cash Crop	0.0165 (0.0149)	0.0162 (0.0178)	0.00493 (0.0130)	-0.0113 (0.0160)
First Year Farming	0.00125*** (0.000450)	0.000827 (0.000500)	0.00116** (0.000434)	0.00128*** (0.000434)
Observations	3,245	3,245	3,227	3,227
R-squared	0.030	0.152	0.059	0.043
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A37: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00140 (0.00340)	-0.00193 (0.00255)		
Customary $\times$ Rental Price	0.0114 (0.00797)	-0.0123*** (0.00337)		
Inverse Travel Time			0.847*** (0.117)	
Customary $\times$ Inverse Travel Time			0.570** (0.220)	
Decrease in Travel Time				2.112*** (0.499)
Customary $\times$ Decrease in Travel Time				0.00511 (0.535)
Residential	0.0893*** (0.0308)	0.0655* (0.0365)	0.0772** (0.0289)	0.0757** (0.0319)
Purchased	0.0894 (0.110)	0.0607 (0.125)	0.127 (0.121)	0.0942 (0.129)
Customary	-0.0294 (0.0642)	-0.0424 (0.0660)	0.0490 (0.0495)	0.190*** (0.0423)
Wealth Index	0.00591*** (0.000731)	0.00340*** (0.00101)	0.00428*** (0.000855)	0.00429*** (0.000979)
Landholdings	0.00159 (0.00146)	0.000698 (0.00102)	0.00211 (0.00132)	0.00190* (0.00110)
Cash Crop	0.142*** (0.0260)	0.0858*** (0.0204)	0.128*** (0.0265)	0.104*** (0.0257)
First Year Farming	0.000702 (0.000487)	0.000614 (0.000561)	0.000603 (0.000477)	0.000674 (0.000491)
Observations	3,244	3,244	3,226	3,226
R-squared	0.082	0.204	0.105	0.098
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A38: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Planted Staple	(3)	(4)
Rental Price	-0.0120** (0.00520)	-0.0111*** (0.00392)		
Customary $\times$ Rental Price	-0.00106 (0.00417)	0.00329 (0.00336)		
Inverse Travel Time			-0.356 (0.273)	
Customary $\times$ Inverse Travel Time			0.00585 (0.217)	
Decrease in Travel Time				-3.102*** (0.641)
Customary $\times$ Decrease in Travel Time				-1.485* (0.747)
Customary	0.0423 (0.0286)	-0.00445 (0.0258)	-0.0578 (0.0657)	-0.140** (0.0667)
Wealth Index	-0.00348** (0.00159)	0.000491 (0.000526)	-0.00292** (0.00137)	2.44e-05 (0.000913)
Landholdings	-0.00312** (0.00124)	-0.00137* (0.000741)	-0.00300** (0.00113)	-0.00368*** (0.000950)
First Year Farming	-1.18e-05 (0.000683)	-0.000878 (0.000788)	2.54e-05 (0.000697)	-0.000149 (0.000696)
Observations	3,254	3,254	3,236	3,236
R-squared	0.060	0.284	0.062	0.105
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A39: OLS results of a dummy equal to one for parcels with at least one plot primarily devoted to a staple crop on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0220 (0.0222)	0.0246 (0.0158)		
Customary $\times$ Rental Price	0.109*** (0.0138)	0.122*** (0.0193)		
Inverse Travel Time			-1.341*** (0.438)	
Customary $\times$ Inverse Travel Time			0.798 (0.492)	
Decrease in Travel Time				-2.162* (1.137)
Customary $\times$ Decrease in Travel Time				0.0894 (1.149)
Customary	0.0934 (0.0770)	0.0746 (0.0747)	-0.503** (0.219)	-0.139 (0.197)
Wealth Index	-0.00164 (0.00230)	-0.00141 (0.00269)	-0.000869 (0.00250)	-0.000146 (0.00236)
Landholdings	-0.00431 (0.00280)	-0.00578 (0.00377)	-0.00522** (0.00249)	-0.00552** (0.00270)
First Year Farming	0.00128* (0.000660)	0.000963 (0.000814)	0.00110 (0.000738)	0.00102 (0.000696)
Observations	3,284	3,284	3,265	3,265
R-squared	0.261	0.380	0.258	0.251
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A40: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0145 (0.00901)	0.00979 (0.00991)		
Customary $\times$ Rental Price	-0.0418*** (0.0118)	-0.0380*** (0.00860)		
Inverse Travel Time			0.956** (0.428)	
Customary $\times$ Inverse Travel Time			0.0188 (0.439)	
Decrease in Travel Time				1.644* (0.888)
Customary $\times$ Decrease in Travel Time				-1.378* (0.747)
Customary	0.0114 (0.0464)	0.0199 (0.0500)	0.275** (0.112)	0.319*** (0.0930)
Wealth Index	-0.000725 (0.00157)	0.00166** (0.000800)	-0.00195 (0.00141)	-0.00128 (0.00129)
Landholdings	0.000974 (0.00144)	0.00173 (0.00128)	0.00133 (0.00138)	0.000802 (0.00153)
First Year Farming	-0.00311*** (0.000849)	-0.00335*** (0.00107)	-0.00309*** (0.000855)	-0.00308*** (0.000824)
Observations	3,250	3,250	3,232	3,232
R-squared	0.047	0.235	0.060	0.054
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A41: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	0.00148 (0.00200)	-0.000819 (0.00109)		
Customary $\times$ Rental Price	0.0161** (0.00610)	0.00299 (0.00197)		
Inverse Travel Time			0.644** (0.261)	
Customary $\times$ Inverse Travel Time			0.323** (0.135)	
Decrease in Travel Time				4.049*** (0.498)
Customary $\times$ Decrease in Travel Time				1.516*** (0.340)
Customary	-0.0898*** (0.0258)	-0.0194 (0.0202)	-0.000372 (0.0461)	0.190*** (0.0521)
Wealth Index	0.00525*** (0.000984)	0.000912 (0.000589)	0.00418*** (0.000932)	0.00111 (0.000741)
Landholdings	0.00194 (0.00146)	0.00134 (0.00102)	0.00261* (0.00153)	0.00316*** (0.00109)
First Year Farming	-0.000385 (0.000598)	-0.000143 (0.000460)	-0.000476 (0.000586)	-0.000259 (0.000493)
Observations	3,248	3,248	3,230	3,230
R-squared	0.105	0.321	0.121	0.199
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A42: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00348 (0.00262)	0.00512*** (0.00179)		
Customary $\times$ Rental Price	0.00912 (0.00873)	0.00419 (0.00261)		
Inverse Travel Time			0.752*** (0.0817)	
Customary $\times$ Inverse Travel Time			0.589*** (0.107)	
Decrease in Travel Time				1.338*** (0.398)
Customary $\times$ Decrease in Travel Time				0.628* (0.317)
Customary	0.00670 (0.0362)	0.00452 (0.0403)	0.0528 (0.0476)	0.0849** (0.0330)
Wealth Index	0.00465*** (0.000696)	0.00203** (0.000876)	0.00325*** (0.000727)	0.00331*** (0.000856)
Landholdings	0.00174** (0.000782)	0.00150* (0.000859)	0.00280*** (0.000913)	0.00262*** (0.000930)
First Year Farming	0.00127*** (0.000466)	0.000888 (0.000529)	0.00121** (0.000466)	0.00135*** (0.000466)
Observations	3,245	3,245	3,227	3,227
R-squared	0.028	0.150	0.057	0.041
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A43: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Pesticides	(3)	(4)
Rental Price	0.000296 (0.00384)	-0.00113 (0.00260)		
Customary $\times$ Rental Price	0.0133 (0.00878)	-0.0126*** (0.00337)		
Inverse Travel Time			0.923*** (0.145)	
Customary $\times$ Inverse Travel Time			0.602*** (0.213)	
Decrease in Travel Time				2.647*** (0.561)
Customary $\times$ Decrease in Travel Time				0.259 (0.449)
Customary	-0.0486 (0.0661)	-0.0465 (0.0661)	0.0440 (0.0536)	0.207*** (0.0460)
Wealth Index	0.00680*** (0.000995)	0.00335*** (0.00102)	0.00492*** (0.00108)	0.00433*** (0.00105)
Landholdings	0.00215 (0.00165)	0.000837 (0.00103)	0.00262* (0.00146)	0.00241** (0.00115)
First Year Farming	0.000359 (0.000491)	0.000475 (0.000540)	0.000286 (0.000467)	0.000436 (0.000493)
Observations	3,244	3,244	3,226	3,226
R-squared	0.057	0.197	0.085	0.085
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A44: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Perennial Crop	(3)	(4)
Rental Price	0.00210 (0.00210)	0.000365 (0.000703)		
Customary $\times$ Rental Price	0.0157** (0.00684)	0.00162 (0.00247)		
Inverse Travel Time			0.552 (0.335)	
Customary $\times$ Inverse Travel Time			0.104 (0.222)	
Decrease in Travel Time				4.653*** (0.651)
Customary $\times$ Decrease in Travel Time				2.108*** (0.661)
Customary	-0.0915*** (0.0281)	-0.00715 (0.0104)	0.0346 (0.0517)	0.196*** (0.0548)
Wealth Index	0.00634*** (0.00200)	0.000476 (0.000675)	0.00540*** (0.00186)	0.00117 (0.00141)
Landholdings	0.00251 (0.00201)	0.000306 (0.000712)	0.00253 (0.00185)	0.00345** (0.00145)
First Year Farming	-0.00260** (0.000989)	-0.00154* (0.000767)	-0.00269** (0.00100)	-0.00243*** (0.000870)
Observations	3,254	3,254	3,236	3,236
R-squared	0.172	0.532	0.183	0.282
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A45: OLS results of a dummy equal to one for parcels with at least one plot primarily devoted to a perennial crop on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C.7.3 Distance to Markets

Changing patterns of input use could be responding not to rising land values, but increasing access to agricultural markets, either for inputs (agroinput dealers) or outputs. However, this should not impact customary parcels differently than freehold, so it seems unlikely to be driving our primary results. However, for a subset of respondents we collected data on the distance to input and output markets, so in this appendix I control for those distances. In tables A46 - A50, I control for the one-way travel cost to the nearest agroinput dealer, to control for access to input markets. In tables A51 - A55, I control for the one-way travel cost to the nearest village in which crops can be sold, focusing on output markets. In results not reported, I also control for the average distance in kilometers at which crops were actually sold in the past season, but when crops were sold at the farm gate this was reported as a zero, making it a difficult to interpret measure (as more remote farms may be more likely to sell at the farm gate). These controls do not change the substance of my results in the main text, indicating access to markets is not driving them; however, I do not use this as a primary control given the 40% reduction in sample size when including these controls.⁵³

⁵³These questions were asked towards the end of the survey, and there was increasing nonresponse towards the end of the multi-hour survey.

VARIABLES	(1)	(2)	(3)	(4)
	Average Insecurity			
Rental Price	0.0250 (0.0182)	0.0281** (0.0127)		
Customary $\times$ Rental Price	0.102*** (0.0155)	0.125*** (0.0234)		
Inverse Travel Time			-1.473*** (0.386)	
Customary $\times$ Inverse Travel Time			0.739 (0.527)	
Decrease in Travel Time				-3.221** (1.338)
Customary $\times$ Decrease in Travel Time				-0.443 (1.238)
Travel Cost to Agroinput Dealer	-1.88e-05* (9.47e-06)	-1.16e-05 (9.67e-06)	-1.15e-05 (8.04e-06)	-1.66e-05* (9.73e-06)
Customary	0.117 (0.0842)	0.0769 (0.0817)	-0.495** (0.213)	-0.169 (0.199)
Wealth Index	-0.00318 (0.00266)	-0.000852 (0.00244)	-0.00221 (0.00253)	-0.000763 (0.00238)
Landholdings	-0.00378 (0.00278)	-0.00678 (0.00419)	-0.00496** (0.00225)	-0.00557** (0.00252)
Cash Crop	0.0686 (0.0712)	0.0227 (0.0820)	0.0920 (0.0776)	0.128 (0.0854)
First Year Farming	0.00156** (0.000748)	0.000932 (0.00104)	0.00148* (0.000770)	0.00138* (0.000735)
Observations	1,724	1,724	1,715	1,715
R-squared	0.274	0.402	0.273	0.268
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A46: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0115 (0.00860)	0.00744 (0.00997)		
Customary $\times$ Rental Price	-0.0420*** (0.0138)	-0.0365*** (0.00887)		
Inverse Travel Time			0.922** (0.396)	
Customary $\times$ Inverse Travel Time			0.0648 (0.515)	
Decrease in Travel Time				1.210 (1.014)
Customary $\times$ Decrease in Travel Time				-1.530 (0.938)
Travel Cost to Agroinput dealer	1.97e-06 (6.81e-06)	-3.89e-06 (7.97e-06)	4.40e-07 (7.70e-06)	-1.51e-06 (7.62e-06)
Customary	0.0211 (0.0412)	0.0134 (0.0496)	0.261* (0.138)	0.296*** (0.104)
Wealth Index	-0.00163 (0.00164)	0.00162* (0.000820)	-0.00280** (0.00135)	-0.00176 (0.00126)
Landholdings	-0.000866 (0.00172)	0.000986 (0.00165)	-0.000336 (0.00167)	-0.000882 (0.00182)
Cash Crop	0.163*** (0.0354)	0.154*** (0.0480)	0.150*** (0.0343)	0.152*** (0.0411)
First Year Farming	-0.00293*** (0.000960)	-0.00363*** (0.00119)	-0.00295*** (0.000968)	-0.00297*** (0.000928)
Observations	1,728	1,728	1,719	1,719
R-squared	0.066	0.272	0.079	0.071
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A47: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	0.000591 (0.00145)	-0.00123 (0.000822)		
Customary $\times$ Rental Price	0.0165*** (0.00610)	0.00688** (0.00270)		
Inverse Travel Time			0.607*** (0.216)	
Customary $\times$ Inverse Travel Time			0.397*** (0.123)	
Decrease in Travel Time				3.767*** (0.508)
Customary $\times$ Decrease in Travel Time				1.621*** (0.410)
Travel Cost to Agroinput Dealer	3.70e-06 (4.11e-06)	1.77e-06 (2.40e-06)	4.00e-06 (4.30e-06)	4.46e-06 (3.89e-06)
Customary	-0.0705** (0.0295)	-0.0124 (0.0258)	-0.0135 (0.0492)	0.159*** (0.0444)
Wealth Index	0.00538*** (0.000993)	0.00173** (0.000747)	0.00443*** (0.000904)	0.00184** (0.000755)
Landholdings	0.000967 (0.00138)	0.00148 (0.00110)	0.00185 (0.00161)	0.00266** (0.00130)
Cash Crop	0.236*** (0.0303)	0.0351 (0.0275)	0.221*** (0.0313)	0.159*** (0.0313)
First Year Farming	0.000577 (0.000513)	0.000531 (0.000466)	0.000479 (0.000534)	0.000591 (0.000485)
Observations	1,727	1,727	1,718	1,718
R-squared	0.207	0.404	0.216	0.271
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A48: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	-0.00500*** (0.00145)	-0.00222* (0.00116)		
Customary $\times$ Rental Price	0.0145 (0.0117)	0.00691** (0.00342)		
Inverse Travel Time			0.817*** (0.0922)	
Customary $\times$ Inverse Travel Time			0.827*** (0.188)	
Decrease in Travel Time				1.477*** (0.464)
Customary $\times$ Decrease in Travel Time				1.122*** (0.408)
Travel Cost to Agroinput Dealer	7.56e-06 (5.61e-06)	1.48e-05*** (4.01e-06)	8.59e-06* (4.94e-06)	8.30e-06 (5.22e-06)
Customary	0.00539 (0.0384)	0.00229 (0.0434)	0.00167 (0.0604)	0.0452 (0.0410)
Wealth Index	0.00570*** (0.000958)	0.00242** (0.00101)	0.00403*** (0.000901)	0.00404*** (0.00108)
Landholdings	0.00115 (0.00146)	0.00106 (0.00120)	0.00235 (0.00157)	0.00231 (0.00159)
Cash Crop	0.0260 (0.0198)	0.0205 (0.0234)	0.0122 (0.0191)	-0.00439 (0.0204)
First Year Farming	0.00125** (0.000542)	0.000757 (0.000559)	0.00116** (0.000528)	0.00132** (0.000537)
Observations	1,725	1,725	1,716	1,716
R-squared	0.041	0.189	0.078	0.055
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A49: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00343 (0.00210)	-0.00614*** (0.00126)		
Customary $\times$ Rental Price	0.0208* (0.0114)	-0.00475 (0.00308)		
Inverse Travel Time			0.929*** (0.155)	
Customary $\times$ Inverse Travel Time			0.895*** (0.321)	
Decrease in Travel Time				2.521*** (0.662)
Customary $\times$ Decrease in Travel Time				0.440 (0.718)
Travel Cost to Agroinput Dealer	1.55e-05*** (5.51e-06)	1.82e-05*** (5.00e-06)	1.67e-05*** (5.22e-06)	1.43e-05** (6.05e-06)
Customary	-0.0451 (0.0815)	-0.0631 (0.0834)	-0.0322 (0.0759)	0.174*** (0.0518)
Wealth Index	0.00750*** (0.000850)	0.00408*** (0.00111)	0.00549*** (0.000880)	0.00525*** (0.00107)
Landholdings	0.00169 (0.00207)	0.000851 (0.00164)	0.00223 (0.00192)	0.00212 (0.00164)
Cash Crop	0.176*** (0.0375)	0.103*** (0.0268)	0.158*** (0.0363)	0.129*** (0.0357)
First Year Farming	0.00121** (0.000537)	0.00129* (0.000648)	0.00106* (0.000560)	0.00119** (0.000554)
Observations	1,724	1,724	1,715	1,715
R-squared	0.114	0.277	0.145	0.135
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A50: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0240 (0.0184)	0.0276** (0.0130)		
Customary $\times$ Rental Price	0.106*** (0.0159)	0.127*** (0.0239)		
Inverse Travel Time			-1.499*** (0.395)	
Customary $\times$ Inverse Travel Time			0.804 (0.519)	
Decrease in Travel Time				-3.289** (1.323)
Customary $\times$ Decrease in Travel Time				-0.212 (1.199)
Travel Cost to Output Market	8.36e-07 (7.03e-06)	-1.30e-06 (4.70e-06)	-1.16e-06 (6.08e-06)	-9.11e-07 (5.97e-06)
Customary	0.116 (0.0843)	0.0744 (0.0821)	-0.522** (0.211)	-0.200 (0.196)
Wealth Index	-0.00290 (0.00259)	-0.000864 (0.00247)	-0.00210 (0.00251)	-0.000627 (0.00237)
Landholdings	-0.00426 (0.00272)	-0.00681 (0.00424)	-0.00521** (0.00222)	-0.00591** (0.00246)
Cash Crop	0.0684 (0.0684)	0.0242 (0.0822)	0.0928 (0.0752)	0.128 (0.0820)
First Year Farming	0.00154** (0.000727)	0.000972 (0.00101)	0.00148* (0.000759)	0.00138* (0.000716)
Observations	1,721	1,721	1,712	1,712
R-squared	0.272	0.402	0.272	0.266
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A51: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0116 (0.00860)	0.00729 (0.0101)		
Customary $\times$ Rental Price	-0.0427*** (0.0132)	-0.0359*** (0.00841)		
Inverse Travel Time			0.918** (0.406)	
Customary $\times$ Inverse Travel Time			0.0683 (0.502)	
Decrease in Travel Time				1.145 (1.021)
Customary $\times$ Decrease in Travel Time				-1.530* (0.898)
Travel Cost to Output Market	-9.23e-07 (1.99e-06)	6.93e-07 (2.01e-06)	7.36e-07 (2.22e-06)	-1.13e-06 (1.94e-06)
Customary	0.0212 (0.0408)	0.0124 (0.0494)	0.260* (0.133)	0.289*** (0.102)
Wealth Index	-0.00169 (0.00162)	0.00164* (0.000832)	-0.00280** (0.00136)	-0.00172 (0.00127)
Landholdings	-0.000814 (0.00166)	0.000966 (0.00168)	-0.000340 (0.00162)	-0.000924 (0.00175)
Cash Crop	0.162*** (0.0355)	0.155*** (0.0477)	0.150*** (0.0349)	0.153*** (0.0411)
First Year Farming	-0.00293*** (0.000956)	-0.00362*** (0.00119)	-0.00297*** (0.000973)	-0.00297*** (0.000931)
Observations	1,725	1,725	1,716	1,716
R-squared	0.066	0.271	0.079	0.071
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A52: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	0.000777 (0.00153)	-0.00109 (0.000793)		
Customary $\times$ Rental Price	0.0167*** (0.00617)	0.00690** (0.00263)		
Inverse Travel Time			0.624*** (0.212)	
Customary $\times$ Inverse Travel Time			0.419*** (0.120)	
Decrease in Travel Time				3.786*** (0.505)
Customary $\times$ Decrease in Travel Time				1.665*** (0.391)
Travel Cost to Output Market	4.44e-06*** (1.29e-06)	2.95e-06** (1.27e-06)	5.89e-06*** (1.41e-06)	6.83e-06*** (1.81e-06)
Customary	-0.0671** (0.0283)	-0.0119 (0.0258)	-0.0106 (0.0469)	0.161*** (0.0425)
Wealth Index	0.00536*** (0.000964)	0.00176** (0.000739)	0.00439*** (0.000890)	0.00183** (0.000765)
Landholdings	0.000985 (0.00137)	0.00148 (0.00108)	0.00188 (0.00160)	0.00270** (0.00130)
Cash Crop	0.233*** (0.0308)	0.0343 (0.0277)	0.216*** (0.0319)	0.154*** (0.0315)
First Year Farming	0.000515 (0.000524)	0.000498 (0.000478)	0.000401 (0.000541)	0.000513 (0.000490)
Observations	1,724	1,724	1,715	1,715
R-squared	0.207	0.398	0.216	0.271
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A53: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	-0.00462*** (0.00157)	-0.00172 (0.00132)		
Customary $\times$ Rental Price	0.0121 (0.0105)	0.00497 (0.00306)		
Inverse Travel Time			0.803*** (0.0975)	
Customary $\times$ Inverse Travel Time			0.761*** (0.195)	
Decrease in Travel Time				1.376*** (0.484)
Customary $\times$ Decrease in Travel Time				0.946** (0.416)
Travel Cost to Output Market	-3.44e-06** (1.60e-06)	-2.72e-06* (1.39e-06)	-1.43e-06 (1.23e-06)	-2.66e-06** (1.23e-06)
Customary	0.00490 (0.0384)	0.00291 (0.0436)	0.0165 (0.0615)	0.0517 (0.0424)
Wealth Index	0.00551*** (0.000922)	0.00255** (0.00103)	0.00394*** (0.000887)	0.00403*** (0.00109)
Landholdings	0.00136 (0.00143)	0.00105 (0.00129)	0.00252 (0.00156)	0.00245 (0.00158)
Cash Crop	0.0275 (0.0192)	0.0215 (0.0228)	0.0132 (0.0186)	-0.000511 (0.0196)
First Year Farming	0.00125** (0.000553)	0.000762 (0.000567)	0.00115** (0.000547)	0.00131** (0.000555)
Observations	1,722	1,722	1,713	1,713
R-squared	0.041	0.185	0.075	0.053
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A54: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00258 (0.00194)	-0.00524*** (0.00138)		
Customary $\times$ Rental Price	0.0186* (0.0106)	-0.00716** (0.00298)		
Inverse Travel Time			0.996*** (0.151)	
Customary $\times$ Inverse Travel Time			0.838** (0.319)	
Decrease in Travel Time				2.652*** (0.660)
Customary $\times$ Decrease in Travel Time				0.312 (0.696)
Travel Cost to Output Market	3.49e-06** (1.34e-06)	5.20e-06*** (1.61e-06)	6.17e-06* (3.09e-06)	4.86e-06** (2.40e-06)
Customary	-0.0422 (0.0810)	-0.0574 (0.0835)	0.00651 (0.0728)	0.203*** (0.0533)
Wealth Index	0.00733*** (0.000857)	0.00402*** (0.00110)	0.00534*** (0.000895)	0.00510*** (0.00108)
Landholdings	0.00203 (0.00209)	0.000930 (0.00167)	0.00257 (0.00189)	0.00241 (0.00163)
Cash Crop	0.173*** (0.0371)	0.0980*** (0.0264)	0.154*** (0.0364)	0.125*** (0.0352)
First Year Farming	0.00114* (0.000574)	0.00112 (0.000696)	0.000974 (0.000597)	0.00111* (0.000592)
Observations	1,721	1,721	1,712	1,712
R-squared	0.110	0.275	0.143	0.133
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A55: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.